

TABULATED DELAYED NEUTRON GROUP PARAMETERS FOR MOLTEN SALT REACTORS

BY

LUKE SEIFERT

DISSERTATION

Submitted in partial fulfillment of the requirements
for the degree of Doctor of Philosophy in Nuclear, Plasma, Radiological Engineering
in the Graduate College of the
University of Illinois at Urbana-Champaign, 2025

Urbana, Illinois

Doctoral Committee:

Professor Kathryn D. Huff, Advisor
More

Abstract

Beta-delayed neutron emitters, or delayed neutron precursors (DNPs), are important for safeguards monitoring and safety considerations in reactor transients, such as shutdown decay-heat, reactor controllability, and reactivity changes. In particular, fluid-fueled molten salt reactors (MSRs) have DNPs which move from areas of high importance in the core to areas of lower importance in the core and ex-core regions. This causes the effective delayed neutron fraction to decrease, leading to changes in reactivity based on the salt flow rate. Many works have investigated how the effective delayed neutron fraction changes based on the movement of fuel salt in MSRs. What these works do not consider, however, is the effect of fuel salt movement and reprocessing on the DNP group parameters used in the transient simulations. This issue is important to resolve, as the DNP group parameters are used for safety-related calculations as well as in safeguards-related detection simulations.

The DNP group parameters for MSRs can be resolved by first generating concentrations that account for the spatially resolved depletion and reprocessing. Shorter lived DNP group parameters are found via pulse irradiation experiments, which occur at very short time scales relative to molten salt flow rates. The longer lived DNP group parameters vary from traditionally calculated values since they are measured with saturation, or infinite, irradiation experiments which occur at time scales several times longer than the longest DNP half life. These longer time scales mean that thermal-hydraulic, thermochemical, and reprocessing effects are able to affect the DNPs.

The concentrations are combined with delayed neutron emission probability and half life data for each nuclide to construct delayed neutron counts as a function of time, a process described in the literature as the "microscopic" approach. A non-linear least-squares solver is then used in order to generate a χ^2 minimized set of DNP group parameters. These updated MSR DNP group parameters are then used in several safety-related transient and safeguards simulations in the Molten Salt Reactor Experiment (MSRE) in order to demonstrate the difference between the group parameters. This work solves an important problem with the DNP group parameters for use in MSRs by providing updated group parameters and a method for generating them dynamically for a given reactor. Tabulated DNP group parameters for various flow rates and energies are provided in this work for future analyses.

Acknowledgments

This research made use of Idaho National Laboratory's High Performance Computing systems located at the Collaborative Computing Center and supported by the Office of Nuclear Energy of the U.S. Department of Energy and the Nuclear Science User Facilities under Contract No. DE-AC07-05ID14517.

Table of Contents

List of Tables	vi
List of Figures	vii
Chapter 1 Introduction	1
Chapter 2 Background and Motivation	3
2.1 Nuclear Safeguards	3
2.2 Safety-Related Reactor Transients	5
2.2.1 Transient Analysis Methods	6
2.3 Delayed Neutron Precursors	13
2.3.1 Overview	13
2.3.2 Measurement	14
2.3.3 Group and Kinetics Parameters	17
2.4 Molten Salt Reactor Depletion	25
2.4.1 Overview	25
2.4.2 Depletion Simulations	27
Chapter 3 Methodology	31
3.1 Tools	31
3.1.1 Python Scripts	31
3.1.2 OpenMC	31
3.1.3 MOOSE Apps	32
3.2 Tabulation	32
3.2.1 Simplified Model	34
3.3 Resolved Delayed Neutron Counts	36
3.3.1 Delayed Neutron Precursor Concentrations	36
3.3.2 Concentrations to Counts	37
3.4 Delayed Neutron Precursor Group Reconstruction	38
Chapter 4 Preliminary Results	40
4.1 Simplified Model Refinement	40
4.1.1 Temperature	40
4.1.2 Target Density	40
4.1.3 Total Irradiation Time	40
4.1.4 Total Decay Time and Decay Time Steps	40
4.2 Tabulation Refinement	40
4.2.1 Large Cycle Time Analysis	41
4.2.2 Reprocessing Rate Scaling	41
4.2.3 Residence Time Variation	41
4.3 Parameter Fit Refinement	41
4.3.1 Time Evaluations	41
4.3.2 Delayed Neutron Count Uncertainties	41

4.3.3	Group Parameter Uncertainties (?)	41
4.4	Group Parameter Comparisons	41
4.4.1	Comparison with Documented Results	41
4.4.2	Comparison of Current Results	41
4.4.3	Comparison with Decay Chain Daughters	41
4.5	Spatially Resolved Depletion Effects on Precursors	41
4.6	Verification	41
4.6.1	Simplified Model	41
4.6.2	No Molten Properties Comparisons	41
4.6.3	Literature Comparison	42
4.7	Validation	42
4.7.1	MSRE Rod Insertion	42
Chapter 5	Proposed Work	43
5.1	Higher-fidelity Model	43
5.1.1	Group Generation	43
5.1.2	Group Evaluation	43
5.1.3	Least-Squares Fission History	43
5.2	DNP Spectra	44
5.3	Resolve DNP spectra	44
5.4	Uncertainty Quantification	44
5.5	Other extensions	44
Appendix A	Appendix A: Derivation of Simplified Flowing Irradiation Delayed Neutron Counts for Equal Residence Times	45
References		48

List of Tables

2.1	Special nuclear material (SNM) classification table recreated from [55].	4
2.2	Half-lives of DNP groups.	22
2.3	Molten Salt Reactor Models Analyzed with Methods Approximating Online Reprocessing from [107] .	26
3.1	MSBR Online Reprocessing Cycle Times from [99]	33

List of Figures

2.1	Experimental setup for measuring delayed neutron emission from irradiation of GODIVA on a sample recreated from [62].	15
2.2	A block diagram of the time-of-flight experimental setup for measuring delayed neutrons from bromine-87 and bromine-88, extracted from the “Democritos” reactor via a fast radiochemical technique recreated from [27].	16
3.1	Flow diagram of simplified model used to generate flow-inclusive precursor group fit parameters. . .	35

Chapter 1

Introduction

- Go through history of group parameters (maybe timeline figure?).
- Discuss MSRs.
- "The next step forward for group parameter development is to have group parameters that accurately reflect the physics that occur in MSRs".
- Introduce the gap [130, 133, 73, 112, 68] that other works do not account for (they investigate movement of DNPs during transient, not in the data-collection stage).
- Discuss methodology at a high level.
- Discuss results at a high level.
- Discuss conclusions of the work at a high level.

Overview of rest of document

1939 Roberts et al. discovery of delayed neutrons [98]. In 1947, Hughes et al. initial use of six groups (kinda) at Argonne heavy water pile [56]. In 1957, Keepin used a sample of U235 in Godiva to generate DNP group parameters (improved group fit). Thus far, only macroscopic approaches have been possible due to insufficient data. For MSRs which have chemical removal of certain elements, tracking individual nuclides is important. In 1989, Brady's thesis used updated data on 271 nuclides identified as DNPs (first attempt at microscopic/summation/aggregate approach). This data was used for ENDF-B/VI was released, and since then there have not been any changes to the DNP group parameters [125, 22]. In 1999, Parish says "Based on the six-group data reported in the literature, it appears that the ENDF/B-VI six-group constants are in need of revision." [91]. In 2014, International Atomic Energy Agency (IAEA) Coordinated Research Project (CRP) on Development of a Reference Database for Beta-delayed neutron emission data [85]. 2020 Leconte uses more than 400 DNPs with data [72].

Brady's thesis has a good history (up to 1988).

This document should contain tabulated data for various fissile nuclides, neutron irradiation energies, in-core residence times, ex-core residence times, and mixing rates. Additionally, data should be provided on the exact concentrations of each nuclide (up to 99% of the DN yield).

Chapter 2

Background and Motivation

This work aims to generate more accurate DNP group parameters and implement those parameters in safeguards and safety-related transient scenarios. Thus, a background is necessary to properly motivate these aspects of this work. Nuclear safeguards and safety-related reactor transients are the primary motivators behind this work, so they are discussed first. Various information relating to MSRs is then provided, including: depletion simulations, DNP treatment, thermochemistry, and the specific reactor analyzed in this work.

2.1 Nuclear Safeguards

Preventing the spread of nuclear material for weapons is the goal of nuclear safeguards. Within nuclear safeguards there are two general categories: proliferation resistance (PR) and physical protection (PP) [95]. This work focuses on PR, which is defined as the nuclear reactor characteristics that impede "...the diversion or undeclared production of nuclear material and the misuse of technology by the Host State seeking to acquire nuclear weapons or other nuclear explosive devices." [95] This definition agrees with the definition made at the IAEA international workshop in 2002 [60].

There are six measures for PR: technical difficulty, cost, time, fissile material type, detection probability, and detection resource efficiency. In this work, the measures analyzed include time and detection probability. The time measure is analyzed by conducting diversion scenarios over various time frames to determine the effects on diversion fingerprints. A very low PR time is 0-3 months, while a very high PR time is >30 years [95]. The detection probability measure goes hand-in-hand with time, as a protracted diversion scenario is likely to have a reduced detection probability. A very low PR detection probability is 0-5 percent, while a very high PR detection probability is 95-100 percent [95].

The Department of Energy (DOE) released an updated nuclear material control and accountability (MCA) Order in February of last year that cancels DOE Order 474.2 Chg 4, which was last updated in 2016 [55]. Based on the more recent Order, SNM is classified as shown in Table 2.1.

Table 2.1: SNM classification table recreated from [55].

Material Type	Accountable Quantity	Weight Field Used for Element	Weight Field Used for Isotope
U-235	1 g	total U	U-235
U-233	1 g	total U	U-233
Pu-242	1 g	total Pu	Pu-242
Pu-239-241	1 g	total Pu	Pu-239 + Pu-241
Pu-238	0.1 g	total Pu	Pu-238
Am-241**	1 g	total Am	Am-241
Am-243**	1 g	total Am	Am-243
Np-237**	1 g	total Am	Np-237

** Americium and neptunium are treated as SNM if they are separated.

Table 2.1 is used along with the graded safeguards table in order to properly classify the attractiveness level and categorization of SNM. These categories include weapons, pure products, high-grade materials, low-grade materials, and all other materials, ranging from attractiveness level "A" to "E". High-grade materials, with attractiveness level "C", include fuel elements and assemblies. Low-grade materials, with attractiveness level "D", include solutions with target concentrations of 1-25 g/L. All other materials, with attractiveness level "E", are made up of highly irradiated forms, solutions with concentrations <1 g/L, or uranium containing <20% uranium-235 or <10% uranium-233.

The quantities provided in Table 2.1 show the minimum reportable, or accountable, quantities for each material type. These quantities make up the lower limit of Category IV materials for "B" through "E" attractiveness level materials. If the material is classified as attractiveness level "C", then any quantity of plutonium or uranium-233 ≥ 6 kg is considered Category I. If the material contained uranium-235, separated neptunium-237, or americium-241 and 243, then an amount ≥ 20 kg is considered Category I.

If the material in the reactor is classified as attractiveness level "D", then any quantity ≥ 16 kg is considered Category II, as Category I quantities do not exist for this attractiveness level. Material of attractiveness level "D" does not have a listed Category I quantity for material containing uranium-235, separated neptunium-237, or americium-241 and 243.

Material of attractiveness level "E" does not have Category I, II, or III quantities. Instead, reportable quantities are considered Category IV.

Significant quantities of material are also of interest, of which there are two categories: direct use and indirect use. Direct use nuclear material consists of ^{233}U , highly enriched uranium (HEU) (^{235}U enriched over 20%), and any amount of ^{239}Pu with less than 80% ^{238}Pu . Indirect use nuclear material consists of thorium and low enriched uranium (LEU) (^{235}U enriched at less than 20%). Significant quantities of the different material categories are as follows [95, 3, 10]:

- Direct use
 - 8kg of Pu
 - 25kg of HEU
 - 8kg of ^{233}U
- Indirect use
 - 75kg of ^{233}U
 - 10t of natural U
 - 20t of depleted U

This information is important for the current work, as it provides values for quantities of diverted material to analyze. The exact classification, materials, and quantities to use will vary based on the specifics of the fuel salt and diversion scenario. However, this information provides a set of data which can be directly referenced to determine the diversion rates.

For example, a fuel salt categorized as direct use with a plutonium diversion would have 8kg of plutonium as a significant quantity. If that fuel salt also had an attractiveness level of "D", then 16kg of plutonium would be Category II. Then, the diversion time could vary over a range of months to years, based on various PR times. Detection of these diversion scenarios is then investigated via change in reactivity due to the change in delayed neutron precursors and fissile concentration. The detection probability is then estimated based on this measurable reactivity change.

2.2 Safety-Related Reactor Transients

An important part of ensuring safe reactor operation is making sure safety-related transient events do not disrupt safe reactor operations. These transients are caused by postulated initiating events (PIEs), which are defined by the IAEA in the IAEA Safety Standard Series No. NS-R-1 [1]. Of particular importance are PIEs which lead to design basis accidents (DBAs). Although No. SSR-2/1 has superseded No. NS-R-1, the PIEs and DBAs for MSRs have not changed [2]. Although there is not specific events listed for MSRs, guidance is provided on what sort of events are of interest.

These events include equipment failures, such as pumps failing or pipes rupturing; human error, such as improper maintenance or faulty operator actions; external events, such as an earthquake; and combinations of events. For combinations of events, the IAEA specifies that these events should make sense in combination and in the time frame modeled. For example, if a particular event lasts a long time or has a long lasting effect, then that event could possibly overlap with another unrelated event. Alternatively, events which are not independent of each other could potentially occur at the same time.

Safety-related transients for MSRs include unprotected loss of heat sink (ULOHS), unprotected loss of flow (ULOF), unprotected transient over power (UTOP), and fuel salt over cooling (FSOC) [76, 30]. An ULOHS transient is when the heat exchanger experiences a failure, possibly caused by a pump failure in the intermediate loop, which causes a loss of cooling of the reactor. For the Molten Salt Fast Reactor (MSFR), Brovchenko et al. have shown that slow ULOHS transients that take place over a minute or longer minimize the increase in fuel salt temperature [21]. An ULOF transient is when the flow rate in the primary loop becomes zero, caused by pump failure in the primary loop. An UTOP transient is a reactivity insertion event, which could be caused by various means, such as removing poisons, adding fissile material, or temperature changes. A FSOC transient is when the fuel salt temperature decreases, which could be caused by an increased heat removal rate from the heat exchanger.

These safety-related transients are important for the current work, as the MSR specific DNP group parameter fits will impact the results of these transients. Depending on the properties of the specific reactor, which affect the group fits, the transient simulation may show that traditional static analyses generate overly or insufficiently conservative results.

2.2.1 Transient Analysis Methods

There are several different methods which have been used for analyzing transients in MSRs. These methods include, as of 2018, "...fluid-flow-velocity field coupled multi-group, time-resolved, neutron diffusion; ... point reactor kinetics schemes; and quasi-statics; generally all coupled, tightly or loosely, to a thermal-hydraulics solver" [126].

Multi-group diffusion

Multi-group diffusion kinetics methods are among the most accurate, and computationally expensive, approaches which can be applied to transient problems. Hurwitz and Ehrlich have a general formulation of multi-group diffusion written in 1953, and many more recent works have used the method [58].

Examples of works that use fluid-flow-velocity field coupled multi-group include GenFOAM and Griffin/Pronghorn coupling using Multiphysics Object-Oriented Simulation Environment (MOOSE) [44, 114]. Other examples can be found in the review by Wooten et al. [66, 45, 65]. Even more works which use the multi-group diffusion method can

be found in the literature [130, 128, 67, 87]

These works typically use Picard iteration with multi-group neutron diffusion coupled with thermal-hydraulics to accurately capture the multi-physics effects of the reactor. However, this accuracy also comes with a higher computational cost than simpler methods. The general form of the coupling can be seen in Equations (2.1) and (2.2) [126].

$$\begin{aligned} \frac{1}{\nu_g} \frac{\partial \phi_g}{\partial t} = & \nabla \cdot D_g \nabla \phi_g + \sum_{g' \neq g}^G \Sigma_{s,g' \rightarrow g} \phi_{g'} \\ & + (1 - \beta) \chi_{p,g} \sum_{g'=1}^G (\nu \Sigma_f)_{g'} \phi_{g'} \\ & \sum_{i=1}^I \chi_{d,g,i} \lambda_i C_i - \Sigma_{a,g} \phi_g - \sum_{g' \neq g}^G \Sigma_{s,g \rightarrow g'} \phi_g \end{aligned} \quad (2.1)$$

$$\frac{\partial C_i}{\partial t} = \beta_i \sum_{g=1}^G (\nu \Sigma_f)_g \phi_g - \lambda_i C_i - \nabla \cdot (C_i \vec{u}) + \nabla \cdot \frac{\nu_T}{Sc_T} \nabla C_i \quad (2.2)$$

The terms in Equations (2.1) and (2.2) are defined as follows:

ν_g is the average number of neutrons produced by fission of group g

ϕ_g is the scalar neutron flux of group g

D_g is the group g diffusion term

G is the total number of energy groups

$\Sigma_{s,g' \rightarrow g}$ is the macroscopic scattering cross section from group g' to group g

β is the delayed neutron fraction

$\chi_{p,g}$ is the number of prompt neutrons born in group g

Σ_f is the macroscopic fission cross section

I is the number of delayed neutron precursor groups

$\chi_{d,g,i}$ is the number of delayed neutrons from precursor group i born in group g

λ_i is the decay constant for group i

C_i is the concentration of group i

$\vec{u}$ is the velocity profile

$\Sigma_{a,g}$ is the macroscopic absorption cross section for group g

β_i is the delayed neutron fraction for group i

ν_T is the eddy viscosity

Sc_T is the turbulent Schmidt number

Modified point kinetics

Point reactor kinetics (PRK) are a simpler approach to modeling a transient, with the basic form given in Equations (2.3) and (2.4). The derivation of the PRK equations can be found in the work by Henry, with a fluid-fueled derivation in the work by Lapenta et al. [53, 70]. Modified point kinetics (MPK) are used for MSRs, as standard PRK do not properly handle the movement of the fuel salt.

$$\frac{dn(t)}{dt} = \frac{\rho(t) - \beta_{eff}}{\Lambda} n(t) + \sum_{i=1}^I \lambda_i C_i(t) \quad (2.3)$$

$$\frac{dC_i(t)}{dt} = \frac{\beta_{eff,i}}{\Lambda} n(t) - \lambda_i C_i(t) \quad (2.4)$$

The terms in Equations (2.3) and (2.4) are defined as follows:

n is the neutron density

ρ is the reactivity

β_{eff} is the effective delayed neutron fraction

Λ is the prompt neutron generation time

λ_i is the group i decay constant

C_i is the concentration of group i

$\beta_{eff,i}$ is the effective delayed neutron fraction from group i

PRK in general struggle to capture spatial effects, such as flow variation, which is important for MSRs. PRK do, however, give a correct prediction for general evolution trends [126]. Wooten et al. provide four different works which use MPK in MSR analysis [49, 83, 111, 129].

The work by Guo et al. analyzes ULOHS and ULOF transients in the Molten Salt Actinide Recycler and Transmuter reactor (MOSART). The PRK equations are modified into the delayed point reactor kinetics (DPK), which model

precursor concentrations in-core and ex-core. These equations, shown in Equations (2.5) through (2.7) (recreated from [49, 111]), are also used by Shi et al. in the Reactor Excursion and Leak Analysis Program (RELAP) for a load demand change, primary flow transient, secondary flow transient, and reactivity transient in the Molten Salt Breeder Reactor (MSBR) [111]. This approach enables modeling of ex-core decay of DNPs, capturing effects at power and for loss of flow, but "...may fail to adequately capture reactivity oscillations in transients occurring at low or zero power and with increases in flow rate" [126].

$$\frac{dn(t)}{dt} = \frac{\rho - \tilde{\beta}_{eff}}{\Lambda} n(t) + \sum_{i=1}^I \lambda_i C_{i,core}(t) \quad (2.5)$$

$$\begin{aligned} \frac{dC_{i,core}(t)}{dt} = & \frac{\beta_i}{\Lambda} n(t) - \left(\lambda_i + \frac{1}{\tau_{core}} \right) C_{i,core}(t) \\ & + \frac{1}{\tau_{loop}} \left(\frac{V_{loop}}{V_{core}} \right) C_{i,loop}(t - \tau_{loop}) \end{aligned} \quad (2.6)$$

$$\frac{dC_{i,loop}(t)}{dt} = \frac{1}{\tau_{core}} \left(\frac{V_{core}}{V_{loop}} \right) C_{i,core}(t) - \left(\lambda_i + \frac{1}{\tau_{loop}} \right) C_{i,loop}(t) \quad (2.7)$$

Intermediate terms used in Equations (2.5) through (2.7) are shown in Equations (2.8) through (2.11).

$$\tilde{\beta}_{eff} = \beta_{static} - \beta_{loss} \quad (2.8)$$

$$\beta_{loss} = \frac{\Lambda(t)}{n(t)} \sum_{i=1}^I \lambda_i C_{i,loop} \quad (2.9)$$

$$\tau_{core} = \frac{M_{core}}{w(t)} \quad (2.10)$$

$$\tau_{loop} = \frac{M_{loop}}{w(t)} \quad (2.11)$$

The terms in Equations (2.5) through (2.11) are defined as follows, with most terms identical to those in the PRK equations:

V_{loop} is the volume of the loop

V_{core} is the volume of the core

β_{static} is the value of β_{eff} without flow

M_{core} is the mass in the core

M_{loop} is the mass in the loop

$w(t)$ is the mass flow rate

A similar method is also used by Suzuki and Shimazu for analyzing a reactivity initiated accident in the FUJI-12 reactor [113]. That work uses a simpler form of the equations, without dividing the precursor concentrations into core and loop regions.

The work by Merle-Lucotte et al. uses a method called "IPK method," which may stand for "Improve Point-Kinetic method," though this is not explicitly stated [83]. This method uses a fixed mesh for neutronics and a mobile mesh for fluid properties. Merle-Lucotte et al. found that the PRK method without any modifications was able to match the IPK method for general trends and end-states, but was unable to capture reactivity oscillations properly. The IPK method was able to capture these oscillations, but assumes "a uniform fission distribution in the core, together with an instantaneous propagation and extraction of the heat produced" [83].

The work by Zanetti et al. creates an extension of the TRACE code, a finite volume code used for light water reactor (LWR) transients [129]. The DNP concentration as a function of space and time is resolved upwind in space and with backward Euler for time. Adjoint-weighted quantities discretized with the trapezoidal rule are then passed to PRK equation, and the set of DNP concentration and the PRK power, or neutron density, equations are resolved using LU factorization.

Another approach to modify the PRK is to alter the kinetics parameters based on a phase-space integration weighted with the adjoint flux [71]. Alternatively, effective flow-inclusive group yields can be calculated, as shown by Haubenreich, and used in PRK equations [52].

Quasi-statics

Quasi-static methods seek to maintain the accuracy of multi-group diffusion approaches while also having the cost savings of PRK. The general approach of these methods is to update PRK parameters with occasional multi-group diffusion simulations to capture changes in flux and DNP spatial distributions [126]. A complete derivation of the quasi-static method can be found in the work by Henry, who discusses the shape and amplitude flux functions, and the work by Devooght, who discusses the generalized quasi-static method [53, 34]. There are also other works which discuss the quasi-static method in detail [11].

Wooten et al. introduce several different works which have used quasi-statics [126]. One of these, from 2005 by Rineiski et al., covers the SIMMER code, which couples neutronics, thermal-hydraulics, and structure codes [97].

The neutron flux is treated as shown in Equation (2.12), where there is a flux amplitude, $N(t)$, which is recalculated more frequently than the flux shape, $\Phi(\vec{x}, t)$. The neutron flux is constrained in the work by a term containing the phase-space integration and the weighting function used.

$$\Phi(\vec{x}, t) = N(t)\psi(\vec{x}, t) \quad (2.12)$$

Rineiski et al. solve "shape" steps to handle the neutronics and DNP spatial concentrations. The equations used in the "shape" steps are recreated here in Equations (2.13) and (2.14). These equations are solved by time-discretizing Equation (2.13) into a set of steady-state-like equations, which are then treated using the DANTSYS module of SIMMER [5].

$$\begin{aligned} \frac{1}{v} \frac{\partial \psi(\vec{x}, t)}{\partial t} + \frac{1}{vN} \frac{dN}{dt} \psi(\vec{x}, t) + M(t)\psi(\vec{x}, t) = \\ (1 - \beta)\chi_p F(t)\psi(\vec{x}, t) + \frac{1}{N} \chi_d \sum_{i=1}^I \lambda_i C_i(\vec{x}, t) + \frac{1}{N} Q(\vec{x}, t) \end{aligned} \quad (2.13)$$

$$\frac{\partial C_i(\vec{x}, t)}{\partial t} = N\beta_i F\psi(\vec{x}, t) - \lambda_i C_i(\vec{x}, t) - \nabla(u(\vec{x}, t)C_i(\vec{x}, t)) \quad (2.14)$$

Within these "shape" steps are also sub-steps referred to as "reactivity" steps. At each of these "reactivity" steps, the PRK parameters are calculated as well as the flux amplitude. The equations used in the "reactivity" steps are recreated here in Equations (2.15) through (2.18).

$$\frac{dN}{dt} = \left(\frac{\rho - \beta_{eff}}{\Lambda} \right) N + q_{eff} + \sum_{i=1}^I \lambda_i (c_{s,i} + c_{m,i}) \quad (2.15)$$

$$\frac{dc_{s,i}}{dt} = \frac{\beta_{eff}}{\Lambda} N - \lambda_i C_{s,i} \quad (2.16)$$

$$\rho = \frac{\langle W(\vec{x}), (\chi F(t) - M(t)) \psi(\vec{x}, t) \rangle}{\langle W(\vec{x}), \chi F(t) \psi(\vec{x}, t) \rangle} \quad (2.17)$$

$$c_{m,i} = \frac{\langle W(\vec{x}), \chi_d C_{m,i} \rangle}{\Lambda \langle W(\vec{x}), \chi F(t) \psi(\vec{x}, t) \rangle} \quad (2.18)$$

Intermediate terms used in Equations (2.15) through (2.18) are shown in Equations (2.19) through (2.20). The $C_s(\vec{x}, t)$ term is computed at steady state by setting the left-hand side of both Equation (2.13) and (2.14) to zero as

well as the fluid velocity to zero at $t=0$. This allows Equation (2.19) to be solved.

$$\frac{\partial C_{m,i}(\vec{x}, t)}{\partial t} = -\nabla \cdot (u(\vec{x}, t) C_{s,i}(\vec{x}, t)) - \lambda_i C_{m,i}(\vec{x}, t) - \nabla \cdot (u(\vec{x}, t) C_{m,i}(\vec{x}, t)) \quad (2.19)$$

$$\frac{\partial C_{s,i}(\vec{x}, t)}{\partial t} = N(t) \beta_i F(t) \psi(\vec{x}, t) - \lambda_i \partial C_{s,i}(\vec{x}, t) \quad (2.20)$$

The terms in Equations (2.12) through (2.20) are defined as follows:

Φ is the neutron flux

$N(t)$ is the neutron flux amplitude

$\psi(\vec{x}, t)$ is the neutron flux shape

v is the average neutron speed

$M(t)$ is the neutron disappearance operator

$F(t)$ is the neutron production operator

β is the delayed neutron fraction

β_{eff} is the effective delayed neutron fraction

χ_p is the prompt neutron energy spectra

χ_d is the delayed neutron energy spectra

χ is the neutron energy spectra

λ_i is the decay constant for DNP group i

$C_i(\vec{x}, t)$ is the concentration of DNP group i

$Q(\vec{x}, t)$ is the external neutron source

$u(\vec{x}, t)$ is the DNP velocity

$C_{s,i}(\vec{x}, t)$ is the "standard" precursor concentration of group i

$C_{m,i}(\vec{x}, t)$ is the "movable" precursor concentration of group i

ρ is the reactivity

$c_{s,i}$ is the PRK "standard" precursor concentration of group i

$c_{m,i}$ is the PRK "movable" precursor concentration of group i

q_{eff} is the PRK external neutron source

Λ is the prompt neutron generation time

$W(x)$ is the weighting function (by default in SIMMER is an adjoint flux computed at steady-state conditions)

The general formulation used for quasi-static methods is very similar among the various works that use it [38, 114].

2.3 Delayed Neutron Precursors

2.3.1 Overview

An important consideration in nuclear reactors, especially in MSRs, are DNPs, discovered in 1939, with initial spectral measurements in 1946 [98, 32, 23]. DNPs are nuclides which decay via β^- decay, producing a daughter, called an emitter, which emits a neutron [9]. In this work, the term DNP will be solely used, as the delay of the neutron emission practically is only due to the beta decay [17, 90]. Classifying between the emitter and the DNP is not relevant for this work.

This delay in the emission of a neutron has a large effect on the reactor period and transient behavior. The time-dependent rate of delayed neutron emission from a single beta-delayed neutron emitting nuclide, or DNP, is shown in Equation (2.21).

$$n_{d,i}(t) = P_{n,i} \lambda_i N_i(t) \quad (2.21)$$

The terms in Equation (2.21) is defined as follows:

- $n_{d,i}$ is the delayed neutron activity, or count rate, from DNP i
- $P_{n,i}$ is the average number of delayed neutrons emitted from a single beta decay of nuclide i , or the emission probability
- λ_i is the decay constant of DNP i
- $N_i(t)$ is the concentration of DNP i

A challenging aspect of treating DNPs in MSRs is the necessary inclusion of a spatial component due to the movement of the fuel salt. In a solid fuel reactor, such as an LWR, a spatial component is not necessary to

properly model the DNPs. Modeling space and time together causes computational costs to increase significantly, making problems involving DNPs in MSRs more challenging than handling DNPs in stationary fuels. Although it is challenging, it is important to properly model the DNPs in reactors.

The DNPs are important to model properly for several reasons. Material accountancy, reactivity worth, decay heat, and reactor controllability (particularly for MSRs) are all key aspects to understand for safe reactor operation [32, 33, 35, 69]. Without properly modeling DNPs, these aspects will not be accurately predicted. This leads to over-conservatism in safety systems and increased costs, difficulties in licensing, and increased risk.

2.3.2 Measurement

There are two approaches used for measurement of DNPs, referred to as the macroscopic and microscopic measurement approaches [91, 105]. The approaches differ initially, as the macroscopic approach uses a fissile sample irradiation to get delayed neutron counts which can be approximated using groups. The microscopic approach instead uses the data from many experiments combined together to simulate a fissile sample irradiation, using the data to generate the delayed neutron counts. This makes the microscopic approach more challenging, as it depends on the work of many experimentalists, though it also offers more detail on the delayed neutron counts.

Macroscopic Measurement of Delayed Neutron Precursors

This section covers the macroscopic approach. Analysis and collection of macroscopic approach data is very common in the literature, as the data from this method is important for reactors [62, 117, 80, 12, 124, 20, 56, 78, 79]. The macroscopic approach entails irradiation of a sample containing a single fissile nuclide at or near a given neutron energy. Small samples are used in order to minimize neutron multiplication within the sample. Two different types of irradiation are used: pulse irradiation, where the irradiation time is shorter than the shortest half-life; and infinite, where the irradiation time is longer than the longest half-life. After irradiation, the delayed neutrons from the sample are counted as a function of time. Equation (2.22) shows the delayed neutron counts as a function of time after irradiation.

$$n_d(t) = \epsilon \sum_{i=1}^I P_{n_i} \lambda_i N_i(t) \quad (2.22)$$

Many of the terms in Equation (2.22) are the same as in (2.21). The terms in Equation (2.22) which are different are defined as follows:

I is the total number of DNPs

ϵ is the detector efficiency

Once this delayed neutron temporal profile is generated, it is common to then use the data to generate group and kinetics parameters for that fissile nuclide and neutron energy. The specifics of the methods used to form those parameters is discussed in Section 2.3.3. Figure 2.1 (recreated from [62]) shows how an experimental setup for collecting delayed neutron counts works.

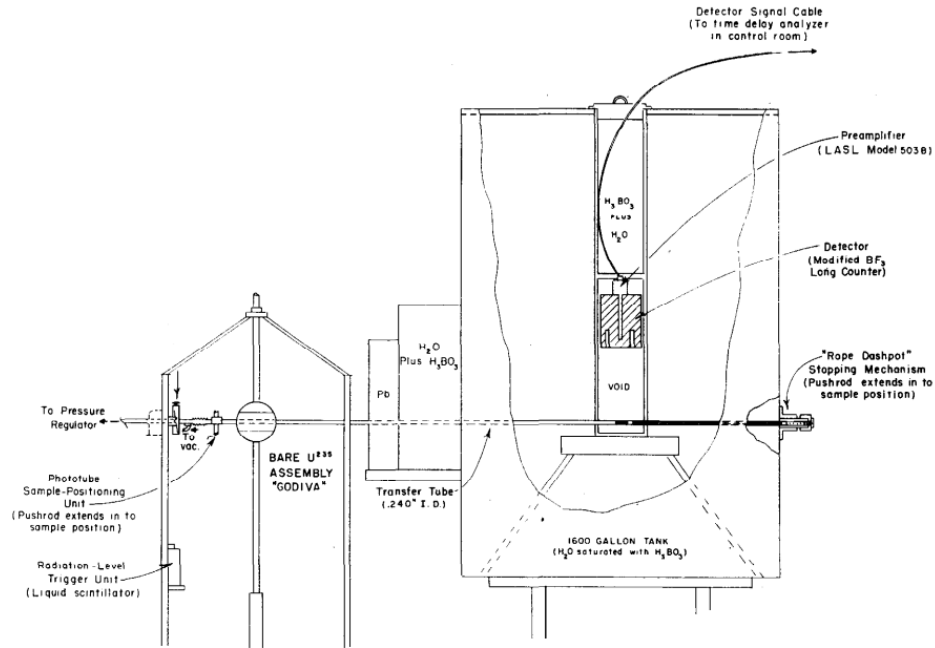


Figure 2.1: Experimental setup for measuring delayed neutron emission from irradiation of GODIVA on a sample recreated from [62].

There are challenges associated with the collection of this experimental data [9]. Experimental noise becomes more significant after long periods of time, as the delayed neutron counts decrease. Additionally, the short lived groups cannot be measured directly due to moderated neutrons from irradiation. This means that both the long and short time-scale measurements have greater uncertainty than the middle time-scale measurements.

Another consideration is measurement of the number of fissions. Although the number of fission events can be measured somewhat accurately using a fission chamber after irradiation or other techniques, this does not account for source multiplication within the irradiated sample. Keepin et al. suggests a technique to correct for the sample self-multiplication, which can cause differences of up to 3% in the correction factor to the number of fissions.

Microscopic Measurement of Delayed Neutron Precursors

The other approach used to measure DNPs is the microscopic approach, investigated in detail in the 1990-2000's [94, 31, 94, 105, 125, 19], with progress through the 2010's via the IAEA CRP on Development of a Reference Database for Beta-delayed neutron emission data [85]. Leconte et al. used this approach as recently as 2020 [72]. This

approach entails taking a sample of a single DNP nuclide and gathering experimental data on its properties. Typically this involves some form of rapid chemical extraction or isotope separator and magnetic tape in order to rapidly measure short lived nuclides.

There are many different experimental methods used to study DNPs, particularly their spectra: on-line mass spectrometry, helium-3 spectrometry, the proton-recoil technique, the time-of-flight method, and the neutron scintillator [32]. On-line mass spectrometry combined with a helium-3 ionization chamber and the proton-recoil technique allows for measurement of the delayed neutron energy spectrum [109, 15]. The time-of-flight method involves measuring the velocity of the delayed neutron by having a known travel distance and time measurement [28, 47, 27]. An example of the time-of-flight experimental setup is shown in Figure 2.2.

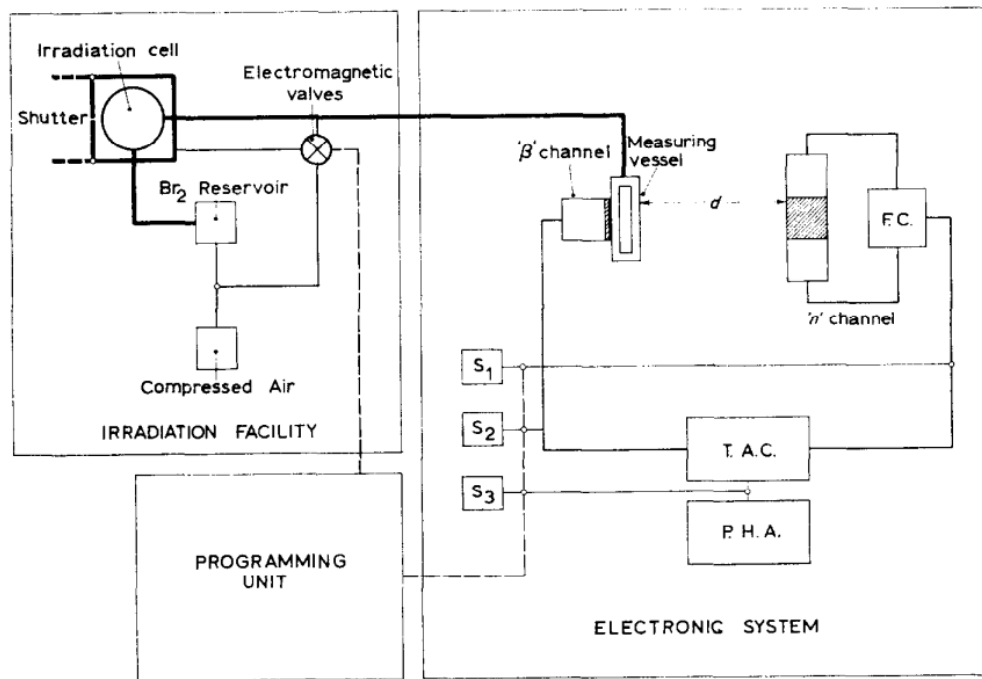


Figure 2.2: A block diagram of the time-of-flight experimental setup for measuring delayed neutrons from bromine-87 and bromine-88, extracted from the “Democritos” reactor via a fast radiochemical technique recreated from [27].

This microscopic measurement data is exceptionally useful, as it allows for treatment of DNPs on a nuclide-by-nuclide basis, rather than macro-scale. This is particularly important for MSRs, where the spatial concentration of DNPs depends on half-lives, yields, thermo-chemistry, and thermal-hydraulics. With data on individual nuclides, higher fidelity treatment of DNPs becomes possible.

There have been many new measurements in the past several decades on these nuclides. Fortunately, the IAEA has worked to compile and evaluate this data into a publicly accessible form [37, 36]. This data, when combined with cumulative fission yields and neutronics data, can be used to model the DNPs explicitly, rather than using

groups. This is particularly useful for understanding what nuclides have the greatest impact as well as for simulating experimental configurations which have yet to be thoroughly investigated for DNP group fitting.

2.3.3 Group and Kinetics Parameters

Overview

In Section 2.3.2, the various methods for experimentally understanding delayed neutron precursors are discussed. The next stage of understanding these precursors is to understand how they are used in kinetics models, as well as the various ways they are represented. The common way of representing the DNP groups is in several distinct groups, typically six. These groups were developed originally for computational efficiency and due to lacking data for individual precursors. Brady and England found that increasing the number of groups for point kinetics calculations caused no significant differences [19]. This shows that merely increasing the number of DNP groups does not provide an increase in accuracy for point kinetics simulations.

The DNP groups come with associated parameters, such as the group decay constants, λ_i , and the group abundances, a_i . These group abundances can be converted to group yields, β_i , by dividing the group abundances by the average number of neutrons emitted per fission event. The group abundances represent the number of delayed neutrons produced per fission in group i , and when summed yield the total delayed neutron yield per fission. The β_i and λ_i terms are the defining parameters for the DNP groups, along with the group spectra, χ_g . With this data, the other group parameters can be derived, such as the total delayed neutron yield, Y , and delayed neutron fraction β .

Although the groups are useful, there are times when individual DNPs nuclides are needed. For example, in 1986, Perry et al. demonstrates that kinetics calculations yield different results when using individual nuclides compared to the six group fit data [93]. In 1989, Brady and England also use the individual nuclides to compare with group data [19], and show that the updated six group parameters offer consistent results with the explicit DNPs. By updating the six group parameters based on the updated data available, greater consistency between experimental results and simulations was made possible.

With the DNPs represented in group form, it also simplifies the calculation of various kinetics parameters. Various kinetics parameters are introduced in Section 2.2.1, though without detail on the purpose of each variable. This section provides more information on what these kinetics and delayed neutron group parameters physically represent, while Sections 2.3.3 and 2.3.3 detail how they are calculated.

Arguably the most well known term in nuclear engineering is the effective multiplication factor, k_{eff} . This term represents the scaling of the number of neutrons over each generation. There is also the infinite multiplication factor, k_{∞} , which represents the multiplication without considering leakage terms. The trend of the effective form

of the term requiring information on the specific reactor geometry continues for the delayed neutron fraction term.

The delayed neutron fraction, β , represents the average number of delayed neutrons emitted per fission event. Alternatively, it can be thought of as representing the ratio of delayed neutron production over all neutron production [84]. This term is directly related to the DNPs, and can be constructed from the DNP groups.

Related to the delayed neutron fraction is the effective delayed neutron fraction, β_{eff} . This term is more sophisticated, as rather than being a physical property of a fissile nuclide, it is dependent on the properties of the reactor core. This term represents the ratio between the adjoint neutron flux and spectrum-weighted delayed-neutron induced fission to the total number of fission events [103]. Alternatively, it can be thought of as the ratio of the spectrum and adjoint weighted delayed neutron production over all spectrum and adjoint weighted neutron production [84]. For MSRs, the movement of the fuel salt also affects the effective delayed neutron fraction, as the integrated adjoint neutron flux is reduced due to movement of the DNPs out of the core and into less neutronically-important regions.

Another term that is important for reactor kinetics is the mean neutron generation time, Λ . The mean neutron generation time is the average "...time between the birth of a neutron and the subsequent absorption that induces fission" [103]. This term also has an effective form, Λ_{eff} , which is the adjoint weighted form. The effective mean neutron generation time can vary significantly, on the order of hundreds of percent, from the mean neutron generation time [119]. In contrast, the effective delayed neutron fraction varies by on the order of 20% from the delayed neutron fraction [119].

The adjoint flux is important for calculating the effective form of many kinetics parameters. The adjoint flux is not a physical quantity, but rather serves as an importance function, describing the importance of the neutron flux in the reactor. For example, areas further from the center of the core are less important than those closer to the center, which is reflected by a decrease in the effective delayed neutron fraction in MSRs as DNPs move away from the center of the core due to fuel salt advection.

Calculation of group parameters

The DNP group parameters are important both for simulation of transients and for their use in calculation of kinetics parameters. The DNP group parameters are formulated based on the fitting of the delayed neutron count rate, as discussed in Section 2.3.2. Equation (2.23) shows the exact formulation for the delayed neutron counts as a function of time measured by a detector for I individual DNPs for any number of fissile nuclides.

$$n_d(t) = \epsilon \sum_{i=1}^I P_{n,i} \lambda_i N_i(t) \quad (2.23)$$

Equation (2.24) instead shows the n group DNP fit measured by the same detector for a pulse irradiation. For

an infinite, or saturation, irradiation, Equation (2.25) is used [20]. These equations are specifically for a single fissile nuclide, and the derivation can be found in the work of Wilson and England [125]. To make this equation generic, the $a_i \lambda_i$ term and a_i term can instead be written as A_i . Additionally, the fission term can be written as $\tilde{F}_s$, to represent either a fission rate or a net number of fissions. These equations are simplifications, as they do not account for decay chain daughter products [29].

$$n_d(t) \approx \epsilon F_s \sum_{i=1}^n a_i \lambda_i e^{-\lambda_i t} \quad (2.24)$$

$$n_d(t) \approx \epsilon \dot{F}_s \sum_{i=1}^n a_i e^{-\lambda_i t} \quad (2.25)$$

The terms in Equations (2.23), (2.24), and (2.25) are defined as follows:

I is the total number of DNPs

n is the number of DNP groups

ϵ is the detector efficiency

$P_{n,i}$ is the emission probability of DNP i

λ_i is the decay constant of the i th DNP or group

$N_i(t)$ is the time dependent number of atoms of nuclide i

F_s is the total number of fission events in the pulse irradiation

$\dot{F}_s$ is the fission rate for a saturation irradiation

a_i is the i th group yield, or group abundance

The pulse irradiation approach gives better statistics on the short lives DNPs, while the longer lived DNPs are better represented by saturation irradiation. Brady discusses this, and uses both pulse and saturation irradiation to form groups [20]. Specifically, Brady uses "...determination of four long-period groups from the infinite irradiation data and four short-period groups from the prompt [pulse] data, later renormalizing the two sets of yields at the second delayed neutron group."

Another benefit of including saturation irradiation data is that it can be used to get the average number of neutrons, prompt and delayed, per fission [29]. This is shown in Equation (2.26).

$$n_d(0) = \epsilon \dot{F}_s \bar{\nu} \quad (2.26)$$

Once the delayed neutron counts are acquired, an over-determined non-linear least squares approach is used to calculate the group abundances and decays. For example, Brady used STEPIT to handle the non-linear least-squares solve with both pulse and infinite irradiation experimental data [20]. Waldo et al. also solve the least-squares problem, though the specific approach used is not specified [124]. The general form of the solve is given in Equation (2.27).

$$A\vec{x} \approx \vec{b} \quad (2.27)$$

Within this least squares equation, the solution vector contains only the group abundances, shown in Equation (2.28).

$$\vec{x} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \quad (2.28)$$

Based on the irradiation type used, the coefficient matrix will change, as shown in Equations (2.24) and (2.25). Equation (2.29) shows the coefficient matrix for a pulse irradiation, though for a saturation irradiation the only change would be removing the λ_i term.

$$A = \begin{pmatrix} \lambda_1 e^{-\lambda_1 t_0} & \lambda_2 e^{-\lambda_2 t_0} & \dots & \lambda_n e^{-\lambda_n t_0} \\ \lambda_1 e^{-\lambda_1 t_1} & \lambda_2 e^{-\lambda_2 t_1} & \dots & \lambda_n e^{-\lambda_n t_1} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_1 e^{-\lambda_1 t_m} & \lambda_2 e^{-\lambda_2 t_m} & \dots & \lambda_n e^{-\lambda_n t_m} \end{pmatrix} \quad (2.29)$$

The target vector contains the delayed neutron data normalized to the number of fissions, or fission rate, and the detector efficiency, as shown in Equation (2.30).

$$\vec{b} = \begin{pmatrix} n_d(t_0)(\epsilon F_s)^{-1} \\ n_d(t_1)(\epsilon F_s)^{-1} \\ \vdots \\ n_d(t_m)(\epsilon F_s)^{-1} \end{pmatrix} \quad (2.30)$$

By solving this non-linear system of equations, the group half lives and abundances can be generated. Once this group data is generated, and if there are multiple fissioning nuclides, k , then Equation (2.31) can be used to determine the total group half lives [32].

$$\lambda_i = \frac{\sum_k \lambda_i^{(k)} \beta_{i,eff}^{(k)}}{\sum_k \beta_{i,eff}^{(k)}} \quad (2.31)$$

These equations thus far have not included the energy spectra, but instead have focused on the delayed neutron count rate. Equation (2.32) shows that including the energy in the delayed neutron count requires the energy spectra, $\chi_i(E)$, of the relevant group to be included. This is also the general form of the saturation and pulse irradiation equations, as it includes A_i , which can represent $a_i \lambda_i$ or a_i depending on the irradiation type. The $\tilde{F}_s$ value also represents that this could be the total number of fissions, for a pulse irradiation, or fission rate, for saturation irradiation.

$$n_d(E, t) \approx \epsilon \tilde{F}_s \sum_{i=1}^n \chi_i(E) A_i e^{-\lambda_i t} \quad (2.32)$$

A similar procedure as discussed for no energy spectra modeling can be used in order to calculate the DNP group energy spectra.

Calculation of group concentrations

Another consideration in calculating group parameters is constructing group concentrations from individual nuclide data. This is less important for group parameters, but is important if there is desire to use the generated group parameters with concentrations which are converted into group concentrations.

Two different methods are considered for the separation and recombination of DNP groups into individual nuclides. These methods have been used for generation of DNP spectral fitting:

- Grouping based on half lives [40, 104]
- Grouping based on fractional contribution to adjacent groups [20]

The grouping based on half lives is easier to implement and understand, as it simply requires comparing each nuclide's half life to a constant value. The groupings based on half lives are given in the top two sets in Table 2.2. The bottom set of half lives is from ENDF/B-VIII.0 and is used with the fractional contribution approach.

Table 2.2: Half-lives of DNP groups.

Source	T_1 [s]	T_2 [s]	T_3 [s]	T_4 [s]	T_5 [s]	T_6 [s]
[40]	$T_{1/2} > 31$	$31 \geq T_{1/2} > 9$	$9 \geq T_{1/2} > 3$	$3 \geq T_{1/2} > 0.8$	$0.8 \geq T_{1/2} > 0.3$	$0.3 \geq T_{1/2}$
[104] [*]	$T_{1/2} > 30$	$30 \geq T_{1/2} > 10$	$10 \geq T_{1/2} > 4$	$4 \geq T_{1/2} > 1.4$	$1.4 \geq T_{1/2} > 0.4$	$0.4 \geq T_{1/2}$
[22]	$T_{1/2} = 51.98$	$T_{1/2} = 21.17$	$T_{1/2} = 5.74$	$T_{1/2} = 2.29$	$T_{1/2} = 0.82$	$T_{1/2} = 0.24$

^{*} Rudstam uses $T_1 = 55.6$, but for the sake of completeness, this has been modified.

The half-life grouping method thus results in a series of comparisons. If a nuclide is within a given group's bounds, the concentration of that nuclide is then added to that group.

The fractional grouping is more complex, but assigns nuclides to groups in a more rigorous manner. The method uses two groups for each nuclide k , which are groups i and $i + 1$. These groups are determined via Equation (2.33). Nuclides which do not have a λ_{i+1} which is larger than λ_k will have the entire fractional contribution to group i .

$$\lambda_i < \lambda_k < \lambda_{i+1} \quad (2.33)$$

The fractional contribution of nuclide k to each group is represented with $f_{k,i}$ and $f_{k,i+1}$. The total fraction must be equal to 1 as shown in Equation (2.34).

$$f_{k,i} + f_{k,i+1} = 1 \quad (2.34)$$

To solve for the fractions of each group each nuclide contributes to, the least squares minimization problem is solved. This is shown in Equation (2.35).

$$\int_0^\infty \left(\lambda_k e^{-\lambda_k t} - f_{k,i} \lambda_i e^{-\lambda_i t} - f_{k,i+1} \lambda_{i+1} e^{-\lambda_{i+1} t} \right)^2 dt \quad (2.35)$$

Solving this equation for where the partial derivatives are zero for each group fraction, combined with Equation (2.34), yields the overdetermined system of equations shown in (2.36).

$$A = \begin{bmatrix} \lambda_i & \frac{2\lambda_i\lambda_{i+1}}{\lambda_i+\lambda_{i+1}} \\ \frac{2\lambda_i\lambda_{i+1}}{\lambda_i+\lambda_{i+1}} & \lambda_{i+1} \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} \frac{2\lambda_i\lambda_k}{\lambda_i+\lambda_k} \\ \frac{2\lambda_k\lambda_{i+1}}{\lambda_k+\lambda_{i+1}} \\ 1 \end{bmatrix}, \quad x = \begin{bmatrix} f_{k,i} \\ f_{k,i+1} \end{bmatrix} \quad (2.36)$$

This system of equations is then solved using the pseudo-inverse, shown in Equation (2.37).

$$x = (A^T A)^{-1} A^T b \quad (2.37)$$

With the fractional contribution vector, the concentration of a given nuclide can then be multiplied by each fraction to show much it contributes to that group. With a running summation, the group concentrations can thus be calculated.

Calculation of kinetics parameters

The other relevant parameters to be calculated are simpler and do not require solving least squares problems. Equation (2.38) shows that to calculate the group yield, the group abundances are divided by the average number of neutrons emitted from each fission event.

$$\beta_i = \frac{a_i}{\bar{\nu}} \quad (2.38)$$

Equation (2.39) shows how the average number of delayed neutrons per fission event, $\bar{\nu}_d$, can be calculated by summing the group abundances [48]. Based on the definition of β_i given in Equation (2.38), it can be seen that the group abundances, a_i , could be rewritten as the group average delayed neutrons per fission event, $\bar{\nu}_{d,i}$.

$$\bar{\nu}_d = \sum_i a_i \quad (2.39)$$

The delayed neutron fraction can be written using the ratio of the average delayed neutrons per fission over the average neutrons per fission. This is shown in Equation (2.40).

$$\beta = \frac{\bar{\nu}_d}{\bar{\nu}} \quad (2.40)$$

The mean neutron generation time can be calculated using Equation (2.41), where l is the prompt neutron

lifetime and k is the multiplication factor. This can be calculated using a time-dependent version of a Monte Carlo code along with PRKs [103].

$$\Lambda = \frac{l}{k} \quad (2.41)$$

Finally, there are the effective parameters: the effective delayed neutron fraction, β_{eff} , and the effective mean neutron generation time, Λ_{eff} . β_{eff} can be calculated explicitly by its definition, which is "...the ratio between the adjoint neutron flux and spectrum-weighted delayed-neutron induced fission to the total number of fission events." This is shown in Equation (2.42) [103].

$$\beta_{eff} = \frac{\int \phi^\dagger \chi_d \beta \nu \Sigma_f \phi \, dE d\vec{\Omega} d\vec{r}}{\int \phi^\dagger \chi \nu \Sigma_f \phi \, dE d\vec{\Omega} d\vec{r}} \quad (2.42)$$

The terms in Equation (2.42) are defined as follows:

$\phi^\dagger$ is the adjoint neutron flux

χ_d is the delayed neutron energy spectrum

χ is the fission neutron energy spectrum

β is the delayed neutron fraction

ν is the average number of neutrons released per fission

Σ_f is the macroscopic fission cross section

ϕ is the neutron flux

Λ_{eff} can also be calculated explicitly by its definition, which is the "...time between the birth of a neutron and the subsequent absorption that induces fission." This is shown in Equation (2.43) [119].

$$\Lambda_{eff} = \frac{\int \phi_0^\dagger \frac{1}{v} \phi_0 \, dE d\vec{\Omega} d\vec{r}}{\int \phi_0^\dagger F_0 \phi_0 \, dE d\vec{\Omega} d\vec{r}} \quad (2.43)$$

The terms in Equation (2.43) are defined as follows:

$\phi_0^\dagger$ is the critical adjoint neutron flux

ϕ_0 is the critical neutron flux

v is the neutron speed

F_0 is the total fission operator

There are other methods used to calculate β_{eff} and Λ_{eff} in the literature [119]. These include the direct integration, or adjoint weighted, method [63]; the eigenvalue, or perturbation, method [53]; and the iterated fission probability method [16]. There is also the pulsed method for calculating Λ_{eff} , and the prompt method for calculating β_{eff} [103]. For β_{eff} , there are also the Meulekamp and Spriggs methods which can be used [84]. Aufiero et al. have conducted a comparison of the Meulekamp, iterated fission probability, and prompt methods, which show that the iterated fission probability and prompt methods both yield similar results while the Meulekamp method is 200 pcm higher for the MSFR [7].

2.4 Molten Salt Reactor Depletion

2.4.1 Overview

There are many different MSR designs in the literature, including [26]: the FUJI series [46], Actinides Molten Salt TransmutER (AMSTER) [120], MOSART [59], Fast Spectrum Molten Salt Reactor (FS-MSR) [54], European MSFR [82], Small Mobile Molten Salt Reactor (SM-MSR) [24], MSBR [99], MSRE [100], Accelerator-Driven Neutron Applications (ADNA) Tier-1 reactor [18], single fluid and two fluid denatured molten salt reactors (DMSRs) [39, 4], and the thorium molten salt reactor (TMSR) [89, 81]. These designs feature a circulating salt containing fissionable material, which has different characteristics than stationary fuels traditionally used in LWRs. In particular, the circulating fuel salt allows for online chemical reprocessing, causes depletion to vary spatially far more than stationary fuel, movement of DNPs, and close coupling of thermal-hydraulics to neutronics.

There are many works in the literature which have investigated online chemical reprocessing of MSRs, as can be seen in Table 2.3 recreated from [107]. Table 2.3 shows which MSR depletion simulations have modeled DNPs in the "DNP" column, which shows most models do not include DNP movement in depletion. The table is sorted by the "Reprocessing" column, which signifies the method(s) used in the different MSR models and is also sorted by the publication year in the "Year" column. The table shows that there are many MSRs, shown in the "Reactor" column, which have had depletion modeled using a variety of spatial refinement levels, shown in the "Model" column, ranging from unit cells to full 3D models.

Table 2.3: Molten Salt Reactor Models Analyzed with Methods Approximating Online Reprocessing from [107]

Publication	Reprocessing	Model	Reactor	DNP	Year
Valdez et al. [61]	Continuous	Unit Cell	N/A	No	2020
Zhuang et al. [132]	Continuous	Full 3D	MSFR	No	2020
Xia et al. [127]	Continuous	Full 3D	N/A	No	2019
Zhou et al. [131]	Continuous	Full 3D	MSBR	Yes	2018
Aufiero et al. [8]	Continuous	Full 3D	MSFR	No	2013
Rodriguez-Vieitez [101]	Continuous	Unit Cell	ADNA	No	2002
Rykhlevskii [106]	Batchwise	Full 3D	MSBR	No	2019
Betzler et al. [14]	Batchwise	Unit Cell	MSBR	No	2017
Ridley and Chvala [96]	Batchwise	Full 3D	Toy	No	2017
Ahmad et al. [4]	Batchwise	Unit Cell	DMSR	No	2015
Park et al. [92]	Batchwise	Full 3D	MSBR	No	2015
Sheu et al. [110]	Batchwise	Full 3D	MOSART	No	2013
Nuttin et al. [88]	Batchwise	Full 3D	MSBR	No	2005
Fiorina et al. [43]	Mixed	2D Slice	MSFR	No	2012
Nagy et al. [86]	Mixed	Unit Cell	Toy	Yes	2008
Merle-Lucotte et al. [81]	Mixed	Full 3D	TMSR	No	2007
Bauman et al. [50]	Mixed	2D Slice	N/A	Yes	1971

Although these works incorporate the effects of online chemical reprocessing on depletion, an effect they do not explicitly model is the spatial effect on depletion. The reason why most works do not include spatial resolution in depletion calculations is due to the computational cost.

2.4.2 Depletion Simulations

Depletion itself is the process of fission, transmutation, decay, reprocessing, and the changes to material compositions this process causes. Simulating depletion requires specific information, such as decay constants, fission yields, and fission and transmutation cross-section data. The fission and transmutation cross-section data can come from a transport calculation, while the other data can be read from a data library [74]. Simulating depletion entails numerically solving the Bateman equations; a generalized form of which is in Equation (2.44).

$$\frac{dN_j}{dt}_{base} = \sum_{i \neq j} [(\gamma_{i \rightarrow j} \sigma_{f,i} \Phi + \lambda_{i \rightarrow j} + \sigma_{i \rightarrow j} \Phi) N_i] - (\lambda_j + \sigma_j \Phi) N_j \quad (2.44)$$

The terms in Equation (2.44) are defined as follows [74]:

N_j is the atomic density of isotope j $\left[\frac{atoms}{cm^3} \right]$

$\gamma_{i \rightarrow j}$ is the fractional fission product yield of j in the fission of isotope i

$\sigma_{f,i}$ is the microscopic fission cross-section of isotope i $[cm^2]$

Φ is the spectrum-averaged scalar flux in the fuel region $\left[\frac{neutrons}{cm^2 s} \right]$

$\lambda_{i \rightarrow j}$ is the decay constant of decay $i \rightarrow j$ $[s^{-1}]$

$\sigma_{i \rightarrow j}$ is the microscopic transmutation cross-section of reaction $i \rightarrow j$ $[cm^2]$

N_i is the atomic density of isotope i $\left[\frac{atoms}{cm^3} \right]$

λ_j is the decay constant of isotope j $[s^{-1}]$

$\lambda_{r,j}$ is the reprocessing constant for removal of isotope j $[s^{-1}]$

σ_j is the microscopic total transmutation cross-section of isotope j $[cm^2]$

$\lambda_{f,j}$ is the feed constant for feed of isotope j $[s^{-1}]$

The matrix form can be seen in (2.45), where N is a vector of compositions, and A is the depletion matrix. This matrix contains the gain and loss terms for each nuclide in N , which vary continuously as a function of time and depend on the concentrations as well. Further documentation on treatment of depletion of these terms can be found in the OpenMC documentation, an open source Monte Carlo transport code which has built-in depletion solvers [102].

$$\frac{dN}{dt} = AN \quad (2.45)$$

There are several approaches to solving this equation, such as the Transmutation Trajectory Analysis (TTA) method and the Chebyshev Rational Approximation Method (CRAM) [75]. Other codes may use external software to compute depletion, such as REM, an in-house code used by Nuttin et al. [88] to implement Fourth order Runge–Kutta. Codes may also use a power series approximation of the matrix exponential, as discussed by Aufiero et al. [8].

Overall, there are many approaches to solving the depletion equation with different computational costs and levels of accuracy. Building a code to handle depletion requires two components: the depletion matrix and a numerical solver.

Stationary fuel

For a stationary fuel, the Bateman equation as shown previously is sufficient to simulate depletion. However, for the purposes of a streamlined comparison with circulating fuel, the Bateman equation can be rewritten to the form shown in Equation (2.46). Over the course of a depletion step, the depletion matrix is assumed to be constant, so the source term varies in time only due to the effects of the changing compositions of other nuclides. This also means that the loss term, $\lambda_{net,j}$, does not vary with time in this equation.

$$\frac{dN_j}{dt} = C(t) - \lambda_{net,j}N_j(t) \quad (2.46)$$

The terms in Equation (2.46) are defined as follows:

C is the source term for nuclide j , containing fission, decay, and transmutation sources

$\lambda_{net,j}$ is the loss term for nuclide j , containing decay and transmutation losses

When solving this equation, the common approach is to have different material blocks in different spatial regions. Each of these material blocks will have different compositions and neutronic properties, such as flux, used for that entire material. This way, no spatial component needs to be considered, and the system of ordinary differential equations can be solved rapidly.

Circulating fuel

Circulating fuel makes the problem more complex, as adding a spatial component means that instead of a system of ordinary differential equations, a system of partial differential equations must be solved. The simplified static

Bateman equation can be modified to include advection and diffusion terms, yielding Equation (2.47). In this equation, the assumption is made that only concentration spatial resolution is applied, so the depletion matrix does not include additional spatial resolution. Thus, only the concentrations and material blocks provide spatial resolution to the problem. More spatial refinement can be provided to the neutronic parameters based on the specific approach used, such as a fully spatially resolved neutron flux.

$$\frac{dN_j}{dt} = C(\vec{r}, t) - \lambda_{net,j} N_j(\vec{r}, t) + \nabla \cdot (D_{mix} \nabla N_j(\vec{r}, t) - \vec{u}(\vec{r}, t) N_j(\vec{r}, t)) \quad (2.47)$$

The terms in Equation (2.46) contain several repeated terms from Equation (2.46), with new terms defined as follows:

D_{mix} is the mixing diffusivity, which sums mass diffusivity and turbulent diffusion

$\vec{u}$ is the fluid velocity field

As previously mentioned, the main challenge faces when solving this problem is the computational cost. However, there are still many works which have attempted to develop methods to get around this problem.

One approach is very computationally inexpensive, and is referred to as the "scaled flux" method [13]. However, this approach does not model the in-core and ex-core regions in a physical manner. Instead, the scaled flux method scales the flux by the fraction of fuel salt in-core compared to ex-core. Another approach is to calculate the steady-state concentration, changing the system of ordinary differential equations to spatial rather than temporal relations [108].

Other approaches, such as Monte Carlo N-Particle (MCNP) and Advanced Dimensional Depletion for Engineering of Reactors (ADDER), or the method proposed by Tano et al. using Griffin and Pronghorn, try to model the flow more explicitly [6, 115]. This approach is more computationally expensive than the other approaches that only require solving a system of ordinary differential equations. However, these works have shown that including advection in depletion directly affects the equilibrium concentration of xenon-135 by roughly 70% [115, 42]. This means that depletion simulations of MSRs need to include spatially resolved depletion to generate accurate results.

However, including spatial resolution in depletion has been referred to as computationally "intractable" [13, 115]. This is because tracking the ~2000 nuclides over space and time requires new methods rather than those used for the time dependent system of ordinary differential equations. Methods for solving the system of partial differential equations require accurately capturing short time scale flow effects, which makes multi-year long depletion simulations exceedingly expensive. Numerical stability of the problem, when using an explicit-time solving scheme, also requires that the Courant-Friedrichs-Lewy (CFL) condition shown in Equation (2.48) to be valid.

$$\Delta t \leq \lambda_{CFL} \frac{\Delta z}{v} \quad (2.48)$$

This equation limits the time step size based on the flow rate and spatial mesh size. Although using a different time solver can work around this condition, the overall computational cost remains high due to the number of nuclides, the time step size, resolution of spatial mesh, and the overall amount of time simulated in depletion.

Chapter 3

Methodology

Maybe I could also add something in the preliminary work on tortuosity/diffusion/turbulent mixing (depending on if it could be implemented into the simplified model). Another consideration could be using a sinusoidal flux shape for the irradiation history (small at inlet, becomes larger, then decreases as it moves towards the outlet) (could be interesting to see if this has a non-negligible effect on the results).

3.1 Tools

To evaluate and generate the updated DNP group parameters, several external software tools are required. In-house Python scripts are used to collect and combine various data, OpenMC generates the simplified group parameters, and the MOOSE app Moltres is used to evaluate those updated group parameters [118, 102, 77]. Further information on each tool is provided in its corresponding sub-section.

3.1.1 Python Scripts

The Python scripts used in the simplified DNP group generation make use of the SciPy, NumPy, Matplotlib, pandas, and Uncertainties packages [122, 51, 57, 116, Eric O. LEBIGOT. A]. These scripts set up the OpenMC sample irradiation simulation, collect data from the IAEA beta-delayed neutron emission database, combine the OpenMC concentrations with the collected half-life and emission probability data, and fit group parameters to the calculated delayed neutron count curve [37].

3.1.2 OpenMC

OpenMC is an open-source Monte Carlo neutron and photon transport code. For the purposes of this work, OpenMC offers many benefits. The Python Application Programming Interface (API) provides useful functionality which makes interfacing with the in-house Python scripts straightforward. Irradiation of the sample is also simple and can be easily configured, adjusting various parameters by altering the Python script inputs. OpenMC also offers useful functionality in depletion, where a zero-power depletion step is rapidly calculated since only depletion is

tracked. The transfer-rates functionality in OpenMC can also be used during depletion, which enables simulation of chemical removal that would occur in an MSR.

For this work, the scripts use the ENDF/B-VIII.0 based depletion chains provided by OpenMC. The thermal spectrum chain is used for neutron irradiation less than 300 eV , while the fast spectrum is used for higher energy neutron irradiations. For the simplified model, monoenergetic neutrons are used. This is because the purpose of the irradiation is to generate group parameters for a given neutron irradiation energy rather than for a particular spectrum.

3.1.3 MOOSE Apps

MOOSE is an open-source, parallel finite element framework for which many applications have been built to handle different physics. Coupling of these physics is simplified using MOOSE, and there are many pre-built applications available to handle a variety of physical phenomena.

Moltres

Moltres is an open-source MOOSE application which can be used for simulation of MSRs. Particularly, this work uses Moltres to simulate the MSRE with different DNP group parameters. Moltres enables a validation study by comparing the results of experimental results with different DNP group parameters. The validation study demonstrates the efficacy of these updated DNP group parameters compared to static group parameters.

3.2 Tabulation

To create tabulated DNP group parameter data in a reactor agnostic manner, the manner of tabulation needs to be reactor agnostic as well. At the same time, a large number of tabulation parameters to fully detail the flow regime is not computationally feasible. Instead, the flow-related tabulation uses only three parameters: a mixing rate (λ_{mix}), an in-core average residence time (τ_{in}), and an ex-core residence time (τ_{ex}).

This approach captures the advection of the fuel salt around the primary loop as well as an approximation of turbulent and diffusive mixing. The terms are based on the averaged behavior from a control volume of fuel salt, so thermal-hydraulic effects such as vortices would be treated as changes in residence times and mixing rate. This method allows an approximate fitting for any circulating fuel reactor geometry, where all that is required is generating the relevant parameters for accessing the tabulated data.

The other tabulation required is for the reprocessing, which can vary in which fission products are removed and their reprocessing rates. Refueling is not considered in this work, as for the short time frames analyzed, the effects

of adding additional fuel are considered to be negligible. Additionally, since the saturation irradiation time frame is on the order of multiple half lives of the longest lived DNP, this means a maximum irradiation time of five minutes is required to reach equilibrium DNPs concentrations. This means that elements which have cycle times on the order of months to years are likely to have negligible changes and do not need to be treated. Elements with changes on the order of seconds to days will have a sufficient contribution such that their inclusion is needed to generate accurate data.

The MSBR reprocessing scheme is used as a base in this work, shown in Table 3.1. The rare earths, seminoble metals, volatile fluorides, discard, and higher nuclides are likely negligible for the time scales studied in this work. However, to ensure that they are negligible, the effects of these larger cycle time terms are studied in Section 4.2.1. Additionally, to better understand how the reprocessing rates affect the group parameters, the cycle times are scaled up and down in Section 4.2.2.

Table 3.1: MSBR Online Reprocessing Cycle Times from [99]

Reprocessing Group	Element(s)	Cycle Time (Full Power)
Rare Earths	Y, La, Ce, Pr, Nd, Pm, Sm, Gd	50 days
Rare Earths	Eu	500 days
Noble Metals	Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te	20 seconds
Seminoble Metals	Zr, Cd, In, Sn	200 days
Gases	Kr, Xe	20 seconds
Volatile Fluorides	Br, I	60 days
Discard	Rb, Sr, Cs, Ba	3435 days
Salt Discard	Th, Li, Be, F	3435 days
Protactinium	Pa	3 days
Higher Nuclides	Np, Pu	16 years

Overall, the tabulations which are used are incident neutron energy, target fissile nuclide, in-core residence time, ex-core residence time, mixing rate, reprocessing scheme, and reprocessing scaling vector. Because each unique tabulation is an extra dimension of data that must be collected, it becomes increasingly expensive to compute all of

these tabulated results. In this work, the reprocessing scheme uses the MSBR reprocessing scheme and a single scaling scalar, rather than a vector containing scaling values for each element. Additionally, for the simplified model, discussed in Section 3.2.1, the mixing rate is not used, which also reduces the net computational cost of generating the tabulated data. The incident neutron energies used are 0.0253 *eV* and 1 *MeV*. For the residence times, a study is carried out in Section 4.2.3 to determine the minimum number of residence times and their values to capture general trends.

3.2.1 Simplified Model

To generate sufficient data for refined tabulation, a simplified model is necessary. This simplified model will also provide estimates for the magnitude of variation which can be expected, as well as general trends. The fission history function is a square-wave, where the fission rate is some positive constant for the in-core residence time and zero for the ex-core residence time. This approximation simplifies the analytical solve required to create the non-linear least-squares problem, as shown in Appendix A.

Figure 3.1 shows how this fission history is used in OpenMC in order to mimic the irradiation on a control volume of fuel salt due to advection. The model takes in eight arguments to pass into OpenMC [102]: the in-core residence time, ex-core residence time, neutron irradiation energy, target fissile nuclide, target density, total irradiation time, temperature, and reprocessing terms. Of these terms, three, shown in green, are not related to tabulation: the temperature, target density, and total irradiation time. The other inputs are in yellow and are tabulated parameters. More detail on this block is provided in Section 3.3.

After OpenMC finishes the fixed source simulation, the final concentrations of nuclides in the irradiated sample are saved for analysis in custom scripts. These scripts take in the concentrations and combine them with the IAEA beta-delayed neutron emission database data to generate a delayed neutron count rate function [37]. A non-linear least squares fitting function from SciPy is then used to generate the half life and group abundances [123]. These results are added to tabulated results for a specific nuclide, neutron energy, in-core residence time, ex-core residence time, reprocessing scheme, and reprocessing scalar. More detail on this block is provided in Section 3.4.

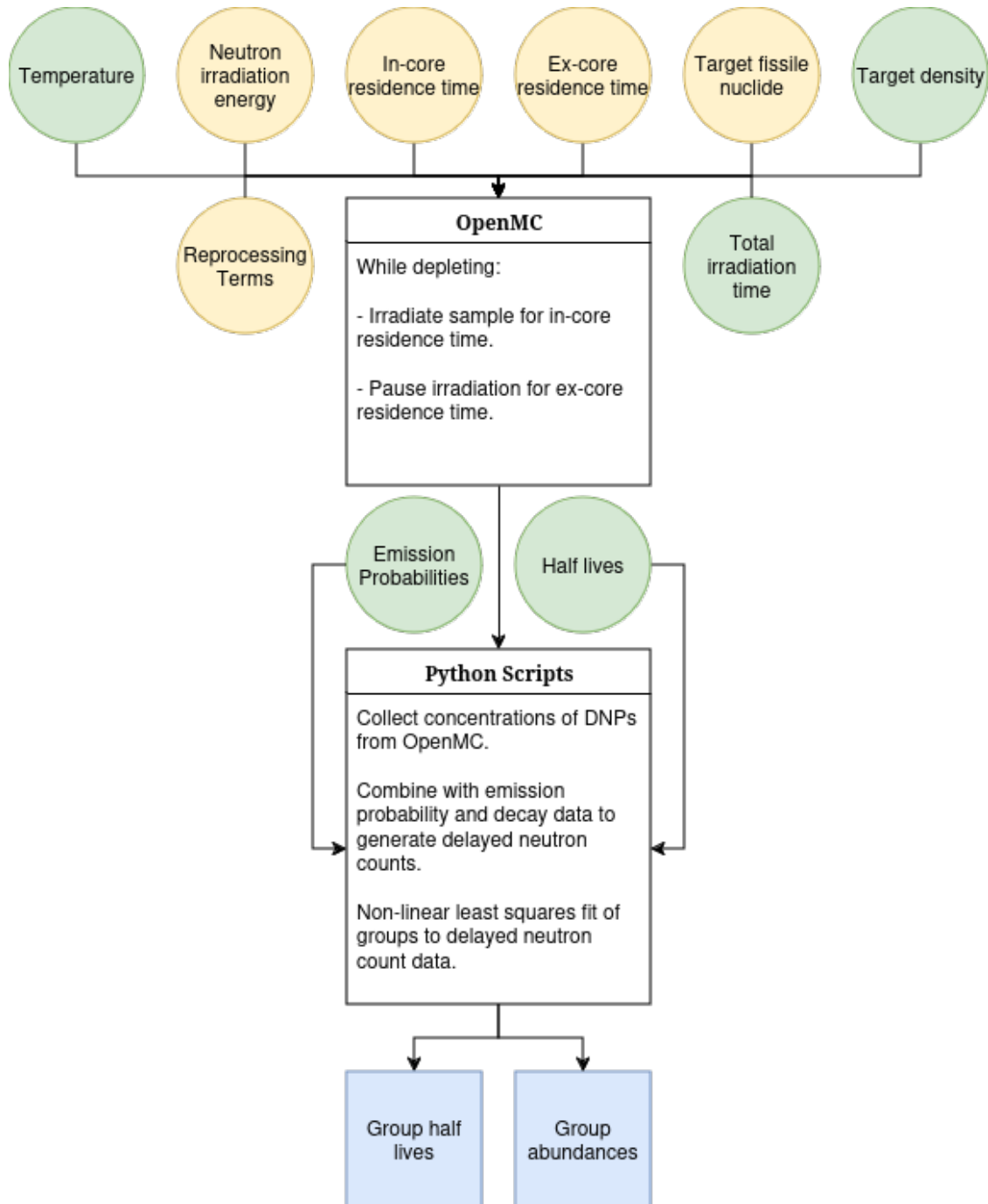


Figure 3.1: Flow diagram of simplified model used to generate flow-inclusive precursor group fit parameters.

There are several considerations required for the OpenMC simulations. The non-tabulated input parameters require some amount of consideration, which is discussed in Section 4.1. In particular, self-shielding of the sample due to density while avoiding self-multiplication, variations in cross sections due to temperature, and under or over irradiation can all affect the results and must be considered [62].

The general form of the OpenMC model is a three gram spherical sample with vacuum boundary conditions. This is treated as a fixed-source problem, with a uniform neutron source distributed evenly throughout the spherical sample. This is not physical, as in reality the neutron source would have a spectrum and would be external. However, for the simplified model, the desired effects to capture are the differences between the group parameters with flow and reprocessing compared to the static results. This means such an approximation does not hinder the effectiveness of this simplified model.

Another consideration for the OpenMC model is having a consistent source particle rate. This is not strictly necessary for group parameter generation, as the values are per-fission, but it is convenient for data analysis, as the tallies are based on the source particles. Equation (3.1) shows how the source rate is calculated to maintain the same source rate between different simulations.

$$\dot{S} = S_{net} \frac{\tau_{in} + \tau_{ex}}{\tau_{in} T} \quad (3.1)$$

The terms in Equation (3.1) are defined as follows:

$\dot{S}$ is the source particle rate

S_{net} is the total number of source particles

τ_{in} is the in-core residence time

τ_{ex} is the ex-core residence time

T is the net irradiation time

3.3 Resolved Delayed Neutron Counts

Although the overview of the simplified model has already been discussed, details on the execution were not. In the following sub-sections, specific information is provided on the methodology behind the OpenMC block. This includes a deeper discussion on how the concentrations are generated from OpenMC, as well as the data which is used with those concentrations to generate a delayed neutron count rate profile.

3.3.1 Delayed Neutron Precursor Concentrations

The OpenMC model uses a fixed-source problem on a three gram spherical sample of a given fissile target. A uniformly distributed source is placed inside the target, and a vacuum boundary condition is used at the edge of the sphere. The sample must have a sufficient volume such that good statistics are achieved, but a volume that is too

large can increase self-shielding and/or self-multiplication. This can be adjusted by modifying the density, where a more dense target increases self-shielding and self-multiplication. A less dense target reduces the probability of fission, which worsens the statistics for the solve.

The fixed-source used in this model is made up entirely of neutrons at a given energy, rather than a spectrum. This energy affects the fission cross sections, transmutation cross sections, and fission yields. Traditionally, tabulated group parameter data is formatted for a single energy or a range of energies. However, the macroscopic experiments, discussed in Section 2.3.2, are performed on a reactor which has a neutron energy spectrum, not mono-energetic neutrons. This could be worked around by modeling a reactor in OpenMC and using a k-eigenvalue solve to generate a neutron energy spectra. However, for this work, the effects of the spectra are considered to be sufficiently small in scale that they do not need to be included. There are also photons which can generate delayed neutrons through photofission, though these are not investigated in this work [64].

The distribution of the neutron source does not precisely describe the actual experimental process, where neutrons irradiate the sample from around the sample rather than inside. However, having the source evenly distributed removes concerns of self-shielding from the problem. The exact spatial locations where fission events occur within the sample should cause negligible changes in the concentrations of nuclides, which is the output from the OpenMC simulation. Assuming a density of $10 \frac{g}{cc}$, then the spherical sample has a radius of approximately 0.4 cm . The spatial resolution of the concentrations in such a small sample should not be significant. Additionally, the concentrations extracted from the OpenMC simulation are spatially collapsed in the post-processing, so the spatial effect of a uniformly distributed source serves only to reduce any self-shielding effects which could reduce the statistics and cause more transmutation than expected.

After running the OpenMC simulation, the concentrations are collected from the depletion results for the final time step of irradiation. This final time step of concentrations can then be passed along to the other scripts. These other scripts handle conversion of concentrations to delayed neutron counts.

3.3.2 Concentrations to Counts

After receiving the concentrations from OpenMC, the concentrations of each nuclide as a function of time for the measurement period can be determined. The concentrations are calculated as a function of half life, as shown in Equation (3.2).

$$N(t) = N_0 e^{-\lambda t} \quad (3.2)$$

This is a simple equation which does not take into account decay chains between nuclides. An alternative

approach which does account for these decay chains is to use repeated decay simulations of OpenMC in order to generate concentrations as a function of time. These concentrations can be used at coarse time steps and linearly interpolated using SciPy, enabling the scripts to access these more accurate concentrations at any time after irradiation of the sample [121]. This approach does not account for particles generated during decay, as it is considered beyond the scope of this work [25]. A comparison of these two approaches is investigated in Section 4.4.3.

The concentrations are combined with Equation (2.23), recreated here as Equation (3.3). The efficiency term, ϵ , is set to 1 in this work, since the simulated method provides a count rate that includes every produced delayed neutron.

$$n_d(t) = \epsilon \sum_{i=1}^I P_{n,i} \lambda_i N_i(t) \quad (3.3)$$

Each nuclide in the IAEA beta-delayed neutron emitter database is iterated through [37]. The delayed neutron count contribution over time is summed for each nuclide, generating the delayed neutron count rate.

3.4 Delayed Neutron Precursor Group Reconstruction

Once the delayed neutron counts as a function of time are generated, they can be converted into DNP group parameters. The process of fitting the group parameters requires a non-linear least squares method, as both the group half lives and group abundances must be determined. The equation describing the problem is given in Equation (2.24), recreated here.

$$n_d(t) \approx \epsilon F_s \sum_{i=1}^n a_i \lambda_i e^{-\lambda_i t}$$

An important consideration for this process is the time values used when generating the group fits. An increased number of times means increased computational cost, but too few times results in a poorer fit [91]. Additionally, the times can be spaced in differing manners, such as linearly or geometrically. This is investigated in Section 4.3.1.

Another consideration for the non-linear least squares solve is the covariance, which is larger for the pulse irradiation fit than it is for the saturation irradiation fit. This is due to the multiplication of the group yield by the group decay constant. Unfortunately, attempts to replace the multiplication of these two values with a new, independent term results in a poor fit. This is because the multiplication of the terms is not independent and is necessary to properly model the fit.

There is also linearization and normalization of the data in the non-linear least squares. The normalization is

handled by taking the delayed neutron counts, dividing by the fission term ¹, and then taking the natural logarithm. These normalized counts are then passed into the non-linear least squares fitting. The linearization is handled by taking the natural logarithm of the entire delayed neutron count fit function. This treatment reduces both the covariance and the χ^2 of the resulting non-linear least squares fit. Bounds are also set from zero to 1000 for each term to speed up the process, while the tolerances for convergence are set to 2.23×10^{16} to ensure the converged solution is well-refined.

The sample self-multiplication, mentioned in Section 2.3.2, is also a possible factor. However, since the number of fissions in the sample are taken directly from the OpenMC tallies, this does not become an issue which must be addressed.

Uncertainties are tracked in this work using the uncertainties package [Eric O. LEBIGOT. A]. Uncertainties in the concentrations generated from OpenMC are not accounted for, but uncertainties in the experimentally measured emission probabilities and half lives are. These uncertainties are used in the minimization of χ^2 , shown in Equation (3.4).

$$\chi^2 = \sum_{t=0}^{t_f} \left(\frac{b_t - A_t x_t}{\sigma_t} \right)^2 \quad (3.4)$$

The terms in Equation (3.4) are defined as follows:

t_f is the final time after irradiation, starting at $t = 0$ immediately following irradiation

b_t is the delayed neutron count vector at the row where the time value is t

A_t is the coefficient matrix at the row where the time value is t

x_t is the solution vector at the row where the time value is t

Once the group fits and their uncertainties are generated, the final step to obtain the finalized group parameters is to combine the saturation and pulse irradiation results. The saturation irradiation provides better statistics for longer-lived groups, while the pulse irradiation is better for shorter-lived groups. Brady combines the irradiation results by using four long-period groups for saturation irradiation and four short-period groups for pulse irradiation. The groups are renormalized at the second group, where the two longest lived group half-lives and the ratio of yields use the saturation results [20]. The remainder of the groups use the pulse irradiation results.

For this work, a similar approach is used, though slightly simplified. The group parameters are calculated for pulse and saturation irradiation, after which the two longest lived groups use the saturation irradiation results while the four shortest lived groups use the pulse irradiation results.

¹The fission term is either the net number of fissions or the fission rate, depending on the irradiation type.

Chapter 4

Preliminary Results

For the preliminary results, the tabulation is not fully resolved. Instead, a default approach is utilized, with variations for refining the tabulation. Thermal fission of uranium-235 is used with the MSBR reprocessing scheme with 10 second in-core and 10-second ex-core residence times at a temperature of 298.15 K and a density of 10 grams per centimeter cubed irradiated over seven minutes with linearly spaced time nodes and decay daughter tracking for the group parameter non-linear least squares fit.

4.1 Simplified Model Refinement

4.1.1 Temperature

4.1.2 Target Density

4.1.3 Total Irradiation Time

4.1.4 Total Decay Time and Decay Time Steps

The bulk of the computational cost comes from writing statepoint files; sufficient resolution is required to ensure a smooth decay curve.

4.2 Tabulation Refinement

Discuss effect of varying reprocessing; determine what differences arise from inclusion of larger cycle time elements (if any). (might need to adjust methodology section based on these results). Scale the reprocessing terms differently to see their effects (see if each can be scaled the same way).

4.2.1 Large Cycle Time Analysis

4.2.2 Reprocessing Rate Scaling

4.2.3 Residence Time Variation

4.3 Parameter Fit Refinement

4.3.1 Time Evaluations

4.3.2 Delayed Neutron Count Uncertainties

4.3.3 Group Parameter Uncertainties (?)

4.4 Group Parameter Comparisons

4.4.1 Comparison with Documented Results

4.4.2 Comparison of Current Results

4.4.3 Comparison with Decay Chain Daughters

Use $N(t)$ from SaltProc with linear interpolation.

4.5 Spatially Resolved Depletion Effects on Precursors

Net delayed neutron yield; group abundances; group decays; comparison of most impactful nuclides (?)

Include polynomial fits (figures) showing how group parameters vary based on tabulated parameters (in-core, ex-core, chemical removal scaling, etc.)

4.6 Verification

4.6.1 Simplified Model

4.6.2 No Molten Properties Comparisons

No flow, reprocessing, thermochemistry, spatial depletion should equal expected results.

4.6.3 Literature Comparison

Should match other Griffin/Pronghorn results. Maybe compare with Moltres/Quasimolto MSRE pump start-up/coast-down. Benchmark Sun Myung talked about?

4.7 Validation

4.7.1 MSRE Rod Insertion

Sun Myung integral rod worth experimental results were slightly larger than Moltres results, maybe using updated group parameters would bring it closer?

Chapter 5

Proposed Work

5.1 Higher-fidelity Model

5.1.1 Group Generation

- Moltres can be used to perform depletion (Nathan Glaser's work)
- Can have spatially resolved irradiation using this functionality (while accounting for thermal-hydraulics)
- Flow analyses can be conducted (channel flow, open core, different fuel diffusion coefficients)
- Verify if simplified model can be used to generate group parameters (if so, when; else why not) (reprocessing from ex-core vs OpenMC dimensionless reprocessing)
- Compare computational cost to accuracy

5.1.2 Group Evaluation

- Moltres takes in group parameters to evaluate reactor transients
- Various safety and safeguards scenarios can be investigated
- Scenarios where the difference between static and flowing group fits leads to incorrect modeling can be discussed

5.1.3 Least-Squares Fission History

See Appendix A. This will probably need numerical evaluation of integral with a polynomial fit afterwards. Could also include $R(t)$ using $N(t)$ rather than assuming exponential decay.

5.2 DNP Spectra

- [32] discusses why it's important in section 6
- Discuss how spectra could be resolved
- Can also be compared between higher fidelity model

5.3 Resolve DNP spectra

5.4 Uncertainty Quantification

Compare with Kozlowski paper Using sensitivity to flow and reprocessing. Uncertainty in half lives and emission probabilities is small. Neglect uncertainty in concentrations from OpenMC (cross sections, fission yields, half lives, transmutations, decay yields, etc.)?

5.5 Other extensions

Diffusion/mixing effect approximated in simplified model? Photoneutrons - investigate Spatially resolved flux - consider, Specific neutron spectra (mentioned in methodology) Thermochemical effects - different elements will have different chemical removal rates, which will affect results (Sam knows more about this and could help me figure out if there would be something here worth investigating)

Appendix A

Appendix A: Derivation of Simplified Flowing Irradiation Delayed Neutron Counts for Equal Residence Times

This derivation begins with the Equations (AA.1) and (AA.2) provided by Wilson and England [125]:

$$R(t) = \sum_{i=1}^n a_i \lambda_i e^{-\lambda_i t} \quad (\text{AA.1})$$

$$n_d(T, t) = \bar{\nu}_d \int_0^T F(\tau) R(T - \tau + t) d\tau \quad (\text{AA.2})$$

The terms in Equation (AA.1) are discussed in Section 2.3.3, and the terms not discussed in the text in Equation (AA.2) are defined as follows:

- T is the length of the fission history
- t is the decay time from the end of the fission history
- $F(\tau)$ is the fission history (in fissions per second)
- $\bar{\nu}_d$ is the average number of delayed neutrons per fission event

Because the results are normalized to a single delayed neutron, the $\bar{\nu}_d$ term is divided out.

To derive an analytical solution to this problem for the simplified approach where the in-core and ex-core residence times, τ_{in} and τ_{ex} , are equivalent, an approximation of the fission history must be made. For this derivation, τ_{res} will be used to represent both the in-core and ex-core residence times. First, the fission history is split into a shape and amplitude function, shown in Equation (AA.3). This essentially just simplifies any further equations such that only the shape needs to be considered.

$$F(t) = F_0 \psi(t) \quad (\text{AA.3})$$

Because the simplified model's fission history is a positive square wave, analytical integration is not feasible. Instead, a Fourier decomposition of the square wave into a sum of sine waves makes the problem far more approachable. For numerical integration this is not an issue, but an analytical solution is useful for setting up the least squares matrices.

$$\psi(t) = \frac{1}{2} + \frac{2}{\pi} \left(\sin\left(\frac{\pi t}{\tau_{res}}\right) + \frac{1}{3} \sin\left(\frac{3\pi t}{\tau_{res}}\right) + \dots \right) \quad (\text{AA.4})$$

Plugging this in and separating the integral gives Equation (AA.5) for group i and for k Fourier terms.

$$\begin{aligned} n_{d,i}(T, t) &= \frac{F_0 a_i \lambda_i}{2} \int_0^T e^{-\lambda_i(T-\tau+t)} d\tau \\ &+ \frac{2F_0 a_i \lambda_i}{\pi} \int_0^T \sin\left(\frac{\pi \tau}{\tau_{res}}\right) e^{-\lambda_i(T-\tau+t)} d\tau \\ &+ \dots \\ &+ \frac{2F_0 a_i \lambda_i}{(2k+1)\pi} \int_0^T \sin\left(\frac{(2k+1)\pi \tau}{\tau_{res}}\right) e^{-\lambda_i(T-\tau+t)} d\tau \end{aligned} \quad (\text{AA.5})$$

Solving the integrals and simplifying yields Equation (AA.6), where each integrated Fourier term generates four terms added to the sum.

$$\begin{aligned} n_{d,i}(T, t) &= \frac{F_0 a_i e^{-\lambda_i t} (1 - e^{-\lambda_i T})}{2} \\ &+ \dots \\ &+ \frac{2F_0 a_i \lambda_i^2 \tau_{res}^2 e^{-\lambda_i t}}{((2k+1)\pi) ((2\pi k + \pi)^2 + \lambda_i^2 \tau_{res}^2)} \sin\left(\frac{\pi(2k+1)T}{\tau_{res}}\right) \\ &+ \frac{2F_0 a_i \lambda_i \tau_{res} e^{-\lambda_i t} e^{-\lambda_i T}}{((2\pi k + \pi)^2 + \lambda_i^2 \tau_{res}^2)} \cos\left(\frac{\pi(2k+1)T}{\tau_{res}}\right) \\ &+ \frac{2F_0 a_i \lambda_i \tau_{res} e^{-\lambda_i t} e^{-\lambda_i T}}{((2k+1)\pi) ((2\pi k + \pi)^2 + \lambda_i^2 \tau_{res}^2)} 2\pi k \\ &+ \frac{2F_0 a_i \lambda_i \tau_{res} e^{-\lambda_i t} e^{-\lambda_i T}}{((2k+1)\pi) ((2\pi k + \pi)^2 + \lambda_i^2 \tau_{res}^2)} \pi \end{aligned} \quad (\text{AA.6})$$

Similar to [125], the irradiation time is taken to the limit as it approaches infinity, though only for the non-periodic terms. By irradiating for several half-lives of time for each DNP, where the longest half-life is roughly one minute, then the irradiation time can be taken as infinite. There are two reasons this limit is taken. Firstly, a pulse irradiation with circulation will not provide any unique results when compared to traditional data which already exists. Additionally, this allows the equation to be simplified, as shown in Equation (AA.7). Only the sine term does not go to zero as the irradiation time goes to infinity, so it is the only term significant for saturation irradiation in the simplified model.

$$\begin{aligned}
n_{d,i}(T, t) &= \frac{F_0 a_i e^{-\lambda_i t}}{2} \\
&+ \dots \\
&+ \frac{2F_0 a_i \lambda_i^2 \tau_{res}^2 e^{-\lambda_i t}}{((2k+1)\pi) \left((2\pi k + \pi)^2 + \lambda_i^2 \tau_{res}^2 \right)} \sin \left(\frac{\pi(2k+1)T}{\tau_{res}} \right)
\end{aligned} \tag{AA.7}$$

The final assumption to make is that the irradiation time, T , is an integer multiple of the residence times, τ_{res} . This assumption makes the sine term 0 for every value of k , which means that the result simplifies to (AA.8).

$$n_d(t) = \frac{F_0}{2} \sum_{i=1}^n a_i e^{-\lambda_i t} \tag{AA.8}$$

This is very similar to the saturation irradiation delayed neutron count equation, shown in Equation (AA.9).

$$n_d(t) = F_0 \sum_{i=1}^n a_i e^{-\lambda_i t} \tag{AA.9}$$

Because of the initial assumption made, which was that half the time is in-core and half ex-core, the average fission rate is half of the fission rate scaling term, F_0 . For both of the cases of constant irradiation and half in-core and half ex-core, the fission term in front is simply the average fission rate over the irradiation time.

References

- [1] (2000). *Safety of Nuclear Power Plants: Design*. Number NS-R-1 in Safety Standards Series. INTERNATIONAL ATOMIC ENERGY AGENCY, Vienna.
- [2] (2016). *Safety of Nuclear Power Plants: Design*. Number SSR-2/1 (Rev. 1) in Specific Safety Requirements. INTERNATIONAL ATOMIC ENERGY AGENCY, Vienna.
- [3] (2022). *IAEA Safeguards Glossary*. Number 3 (Rev. 1) in International Nuclear Verification Series. INTERNATIONAL ATOMIC ENERGY AGENCY, Vienna.
- [4] Ahmad, A., McClamrock, E. B., and Glaser, A. (2015). Neutronics calculations for denatured molten salt reactors: Assessing resource requirements and proliferation-risk attributes. *Annals of Nuclear Energy*, 75:261–267.
- [5] Alcouffe, R. E., Baker, R. S., Brinkley, F. W., Marr, D. R., ODell, R. D., and Walters, W. F. (1995). Dantsys: a diffusion accelerated neutral particle transport code system. Technical report, Los Alamos National Lab.(LANL), Los Alamos, NM (United States).
- [6] Anderson, K., Mascolino, V., and Nelson, A. G. (2022). User Guide to the Advanced Dimensional Depletion for Engineering of Reactors (ADDER) Software (V.1.01). Technical Report ANL/RTR/TM-21/8-Rev.1, Argonne National Laboratory (ANL), Argonne, IL (United States).
- [7] Aufiero, M., Brovchenko, M., Cammi, A., Clifford, I., Geoffroy, O., Heuer, D., Laureau, A., Losa, M., Luzzi, L., Merle-Lucotte, E., et al. (2014). Calculating the effective delayed neutron fraction in the molten salt fast reactor: Analytical, deterministic and monte carlo approaches. *Annals of Nuclear Energy*, 65:78–90.
- [8] Aufiero, M., Cammi, A., Fiorina, C., Leppänen, J., Luzzi, L., and Ricotti, M. (2013). An extended version of the SERPENT-2 code to investigate fuel burn-up and core material evolution of the Molten Salt Fast Reactor. *Journal of Nuclear Materials*, 441(1-3):473–486.
- [9] Bank, N. D. (2000). The JEF-2.2 nuclear data library. JEFF report 17.
- [10] Bathke, C. G., Ebbinghaus, B. B., Collins, B. A., Sleasford, B. W., Hase, K. R., Robel, M., Wallace, R. K., Bradley, K. S., Ireland, J. R., Jarvinen, G. D., Johnson, M. W., Prichard, A. W., and Smith, B. W. (2012). The attractiveness of materials in advanced nuclear fuel cycles for various proliferation and theft scenarios. *Nuclear Technology*, 179(1):5–30.
- [11] Becker, M. (1968). A generalized formulation of point nuclear reactor kinetics equations. *Nuclear Science and Engineering*, 31(3):458–464.
- [12] Benedetti, G., Cesana, A., Sangiust, V., Terrani, M., and Sandrelli, G. (1982). Delayed neutron yields from fission of uranium-233, neptunium-237, plutonium-238,-240,-241, and americium-241. *Nuclear Science and Engineering*, 80(3):379–387.
- [13] Betzler, B. (2021). Liquid-fueled Molten Salt Reactor Depletion Modeling. Technical report, Oak Ridge National Lab. (ORNL), Oak Ridge, TN (United States). 2.
- [14] Betzler, B. R., Powers, J. J., and Worrall, A. (2017). Molten salt reactor neutronics and fuel cycle modeling and simulation with SCALE. *Annals of Nuclear Energy*, 101:489–503.

- [15] Beyer, R., Grosse, E., Heidel, K., Hutsch, J., Junghans, A., Klug, J., Legrady, D., Nolte, R., Röttger, S., Sobiella, M., et al. (2007). Proton-recoil detectors for time-of-flight measurements of neutrons with kinetic energies from some tens of keV to a few MeV. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 575(3):449–455.
- [16] Bohl, L., Margolis, S., and Radkowsky, A. (1964). Computation of parameters in the kinetics equations. *Naval Reactors Physics Handbook*, 1:873–877.
- [17] Bohr, N. and Wheeler, J. A. (1939). The mechanism of nuclear fission. *Physical Review*, 56(5):426.
- [18] Bowman, C. D. (2000). Once-through thermal-spectrum accelerator-driven light water reactor waste destruction without reprocessing. *Nuclear Technology*, 132(1):66–93.
- [19] Brady, M. and England, T. (1989). Delayed neutron data and group parameters for 43 fissioning systems. *Nuclear Science and Engineering*, 103(2):129–149.
- [20] Brady, M. C. (1988). *Evaluation and application of delayed neutron precursor data*. PhD thesis, Texas A&M University.
- [21] Brovchenko, M., Heuer, D., Merle-Lucotte, E., Allibert, M., Ghetta, V., Laureau, A., and Rubiolo, P. (2013). Design-related studies for the preliminary safety assessment of the molten salt fast reactor. *Nuclear Science and Engineering*, 175(3):329–339.
- [22] Brown, D., Chadwick, M., Capote, R., Kahler, A., Trkov, A., Herman, M., Sonzogni, A., Danon, Y., Carlson, A., Dunn, M., Smith, D., Hale, G., Arbanas, G., Arcilla, R., Bates, C., Beck, B., Becker, B., Brown, E., Casperson, R., Conlin, J., Cullen, D., Descalle, M.-A., Firestone, R., Gaines, T., Guber, K., Hawari, A., Holmes, J., Johnson, T., Kawano, T., Kiedrowski, B., Koning, A., Kopecky, S., Leal, L., Lestone, J., Lubitz, C., Damián, J. M., Mattoon, C., McCutchan, E., Mughabghab, S., Navratil, P., Neudecker, D., Nobre, G., Noguere, G., Paris, M., Pigni, M., Plompen, A., Pritychenko, B., Pronyaev, V., Roubtsov, D., Rochman, D., Romano, P., Schillebeeckx, P., Simakov, S., Sin, M., Sirakov, I., Sleaford, B., Sobes, V., Soukhovitskii, E., Stetcu, I., Talou, P., Thompson, I., van der Marck, S., Welser-Sherrill, L., Wiarda, D., White, M., Wormald, J., Wright, R., Zerkle, M., Žerovnik, G., and Zhu, Y. (2018). ENDF/B-VIII.0: The 8th major release of the nuclear reaction data library with CIELO-project cross sections, new standards and thermal scattering data. *Nuclear Data Sheets*, 148:1 – 142. Special Issue on Nuclear Reaction Data.
- [23] Burgy, M., Pardue, L., Willard, H., and Wollan, E. (1946). *Energies of the Delayed Neutrons from U235 Fission Products*, volume 16. Technical Information Division, Atomic Energy Commission.
- [24] CASINO, W., Sorensen, K., and Whitener, C. A. (2007). A small mobile molten salt reactor (sm-msr) for underdeveloped countries and remote locations. *Transactions of the American Nuclear Society*, 97:907–908.
- [25] Chen, Y. and Fischer, U. (2002). Rigorous mcnp based shutdown dose rate calculations: computational scheme, verification calculations and application to iter. *Fusion Engineering and Design*, 63:107–114.
- [26] Cheng, M.-S., Dai, Z.-M., et al. (2014). Development of a three dimension multi-physics code for molten salt fast reactor. *Nucl. Sci. Tech.*, 25:010601.
- [27] Chrysochoides, N., Anoussis, J., Mitsonias, C., and Perricos, D. (1971). Measurement of the low energy spectrum of delayed neutrons from 87Br and 88Br precursors. *Journal of Nuclear Energy*, 25(11):551–556.
- [28] Chulick, E., Reeder, P., Bemis Jr, C., and Eichler, E. (1971). Energy spectrum of delayed neutrons from the spontaneous fission of 252Cf. *Nuclear Physics A*, 168(2):250–258.
- [29] Conant, J. F. and Palmedo, P. F. (1971). Measurement of the delayed-neutron fractions for thermal fission of uranium-235, plutonium-239, and uranium-233. *Nuclear Science and Engineering*, 44(2):173–179.
- [30] Daher, D., Kotb, M., Khalaf, A., El-Koliel, M. S., and Soliman, A. Y. (2022). Improvement of a design of the molten salt fast reactor. *International Journal of Advanced Nuclear Reactor Design and Technology*, 4(2):87–100.
- [31] D’Angelo, A. and Rowlands, J. L. (2002). A review of delayed neutron data for calculating effective delayed neutron fractions. In *Proc. PHYSOR*, pages 7–10.

- [32] Das, S. (1993). Importance of delayed neutrons in nuclear research. Technical report, Bhabha Atomic Research Centre.
- [33] Das, S. and Srinivasan, M. (1977). Kinetics of zero energy fast reactor purnima. Technical report, Bhabha Atomic Research Centre.
- [34] Devooght, J. (1980). Quasistatic solutions of reactor kinetics. *Annals of Nuclear Energy*, 7(1):47–58.
- [35] Dickens, J. (1991). Analysis of beta-ray data important to decay heat predictions. *Nuclear Science and Engineering*, 109(1):92–102.
- [36] Dillmann, I., Dimitriou, P., and Singh, B. (2015). Summary report of 2nd research coordination meeting on development of a reference database for beta-delayed neutron emission. Technical report, International Atomic Energy Agency.
- [37] Dimitriou, P., Dillmann, I., Singh, B., Piksaikin, V., Rykaczewski, K., Tain, J., Algora, A., Banerjee, K., Borzov, I., Cano-Ott, D., Chiba, S., Fallot, M., Foligno, D., Grzywacz, R., Huang, X., Marketin, T., Minato, F., Mukherjee, G., Rasco, B., Sonzogni, A., Verpelli, M., Egorov, A., Estienne, M., Giot, L., Gremyachkin, D., Madurga, M., McCutchan, E., Mendoza, E., Mitrofanov, K., Narbonne, M., Romojaro, P., Sanchez-Caballero, A., and Scielzo, N. (2021). Development of a reference database for beta-delayed neutron emission. *Nuclear Data Sheets*, 173:144–238. Special Issue on Nuclear Reaction Data.
- [38] Dulla, S., Ravetto, P., and Rostagno, M. (2004). Neutron kinetics of fluid–fuel systems by the quasi-static method. *Annals of Nuclear Energy*, 31(15):1709–1733.
- [39] Engel, J., Bauman, H., Dearing, J., Grimes, W., McCoy, H., and Rhoades, W. A. (1980). Conceptual design characteristics of a denatured molten-salt reactor with once-through fueling. Technical report, Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States).
- [40] England, T., Wilson, W., Schenter, R., and Mann, F. (1983). Aggregate delayed neutron intensities and spectra using augmented endf/bv precursor data. *Nuclear Science and Engineering*, 85(2):139–155.
- [Eric O. LEBIGOT. A] Eric O. LEBIGOT. A. Uncertainties: a python package for calculations with uncertainties.
- [42] Fei, T., Shahbazi, S., Fang, J., and Shaver, D. (2022). Validation of NEAMS Tools Using MSRE Data. Technical Report ANL/NSE-22/48, Argonne National Laboratory (ANL), Argonne, IL (United States).
- [43] Fiorina, C., Cammi, A., Krepel, J., Mikityuk, K., and Ricotti, M. E. (2012). Preliminary Analysis of the MSFR Fuel Cycle Using Modified-EQL3D Procedure. In *Volume 4: Codes, Standards, Licensing, and Regulatory Issues; Fuel Cycle, Radioactive Waste Management and Decommissioning; Computational Fluid Dynamics (CFD) and Coupled Codes; Instrumentation and Co*, page 293, Anaheim, California, USA. ASME.
- [44] Fiorina, C., Clifford, I., Aufiero, M., and Mikityuk, K. (2015). Gen-foam: a novel openfoam® based multi-physics solver for 2d/3d transient analysis of nuclear reactors. *Nuclear Engineering and Design*, 294:24–37.
- [45] Fiorina, C., Lathouwers, D., Aufiero, M., Cammi, A., Guerrieri, C., Kloosterman, J. L., Luzzi, L., and Ricotti, M. E. (2014). Modelling and analysis of the msfr transient behaviour. *Annals of Nuclear Energy*, 64:485–498.
- [46] Furukawa, K., Lecocq, A., Kato, Y., and Mitachi, K. (1990). Thorium molten-salt nuclear energy synergetics. *Journal of nuclear science and technology*, 27(12):1157–1178.
- [47] Garcia, A., Martínez, T., Cano-Ott, D., Castilla, J., Guerrero, C., Marín, J., Martínez, G., Mendoza, E., Ovejero, M., Reillo, E., et al. (2012). Monster: a time of flight spectrometer for β -delayed neutron emission measurements. *Journal of Instrumentation*, 7(05):C05012.
- [48] Gremyachkin, D., Piksaikin, V., Mitrofanov, K., and Egorov, A. (2015). Verification of the evaluated fission product yields data from the neutron induced fission of ^{235}U , ^{238}U and ^{239}Pu based on the delayed neutron characteristics. *Progress in Nuclear Energy*, 83:13–25.

- [49] Guo, Z., Zhang, D., Xiao, Y., Tian, W., Su, G., and Qiu, S. (2013). Simulations of unprotected loss of heat sink and combination of events accidents for a molten salt reactor. *Annals of Nuclear Energy*, 53:309–319.
- [50] H. F. Bauman, G. W. Cunningham III, J. L. Lucius, H. T. Kerr, and C. W. Craven, Jr. (1971). ROD: a nuclear and fuel-cycle analysis code for circulating-fuel reactors.
- [51] Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M. H., Brett, M., Haldane, A., del Río, J. F., Wiebe, M., Peterson, P., Gérard-Marchant, P., Sheppard, K., Reddy, T., Weckesser, W., Abbasi, H., Gohlke, C., and Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, 585(7825):357–362.
- [52] Haubenreich, P. (1962). Prediction of effective yields of delayed neutrons in msre. Technical report, Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States).
- [53] Henry, A. F. (1958). The application of reactor kinetics to the analysis of experiments. *Nuclear Science and Engineering*, 3(1):52–70.
- [54] Holcomb, D., Flanagan, G., Patton, B., Gehin, J., Howard, R., and Harrison, T. (2011). Fast spectrum molten salt reactor options. *ORNL/TM-2011/105*.
- [55] Holmer, D. (2024). Doe o 474.2a, nuclear material control and accountability.
- [56] Hughes, D., Dabbs, J., Cahn, A., and Hall, D. (1948). Delayed neutrons from fission of u 235. *Physical Review*, 73(2):111.
- [57] Hunter, J. D. (2007). Matplotlib: A 2d graphics environment. *Computing in Science & Engineering*, 9(3):90–95.
- [58] Hurwitz Jr, H. and Ehrlich, R. (1953). Multigroup methods. *AND TECHNOLOGY*, page 81.
- [59] Ignatiev, V., Feynberg, O., Gnidoi, I., Merzlyakov, A., Smirnov, V., Surenkov, A., Tretiakov, I., Zakirov, R., Afonichkin, V., Bovet, A., et al. (2007). Progress in development of li, be, na/f molten salt actinide recycler & transmuter concept. In *Proceedings of ICAPP*, volume 7, pages 13–18.
- [60] International Atomic Energy Agency (2002). Proliferation resistance fundamentals for future nuclear energy systems. In *Proceedings of the IAEA International Workshop on Future Nuclear Energy Systems*, Vienna, Austria. IAEA Department of Safeguards. Technical Report.
- [61] Jr. Vicente Valdez, P., Betzler, B., Wieselquist, W., and Fratoni, M. (2020). Modeling Molten Salt Reactor Fission Product Removal with SCALE. Technical Report ORNL/TM-2019/1418, 1608211, ORNL.
- [62] Keepin, G. R., Wimett, T., and Zeigler, R. (1957). Delayed neutrons from fissionable isotopes of uranium, plutonium and thorium. *Journal of Nuclear Energy (1954)*, 6(1-2):IN2–21.
- [63] Keepin, G. R. G. R. (1965). *Physics of nuclear kinetics*. Addison-Wesley series in nuclear science and engineering. Addison-Wesley.
- [64] Kinlaw, M. T. (2007). *Delayed neutron group parameters from photofission*. PhD thesis, Idaho State University.
- [65] Kópházi, J., Lathouwers, D., and Kloosterman, J. (2009). Development of a three-dimensional time-dependent calculation scheme for molten salt reactors and validation of the measurement data of the molten salt reactor experiment. *Nuclear science and engineering*, 163(2):118–131.
- [66] Křepel, J., Rohde, U., Grundmann, U., and Weiss, F.-P. (2007a). Dyn3d-msr spatial dynamics code for molten salt reactors. *Annals of Nuclear Energy*, 34(6):449–462.
- [67] Křepel, J., Rohde, U., Grundmann, U., and Weiss, F.-P. (2007b). Dyn3d-msr spatial dynamics code for molten salt reactors. *Annals of Nuclear Energy*, 34(6):449–462.
- [68] Kupriyanov, K., Feinberg, O., and Ignatiev, V. (2022). Effective delayed neutron fraction in a molten salt reactor with circulating fuel. *Physics of Atomic Nuclei*, 85(8):1391–1399.

- [69] Lane, J. A., MacPherson, H. G., and Maslan, F. (1958). *Fluid fuel reactors: Molten salt reactors, aqueous homogeneous reactors, fluoride reactors, chloride reactors, liquid metal reactors and why liquid fission*. Addison-Wesley Pub. Co.
- [70] Lapenta, G., Mattioda, F., and Ravetto, P. (2001). Point kinetic model for fluid fuel systems. *Annals of Nuclear Energy*, 28(17):1759–1772.
- [71] Lapenta, G., Ravetto, P., et al. (2000). Basic reactor physics problems in fluid-fuel recirculated reactors. *KERNTECHNIK-BILINGUAL EDITION*-, 65(5/6):250–253.
- [72] Leconte, P., Archier, P., De Saint Jean, C., Diniz, R., Dos Santos, A., Fautrat, L., Foligno, D., Geslot, B., Gilad, E., Tamagno, P., et al. (2020). New delayed neutron group constants and covariances for lwr core applications, combining summation calculations and integral experiments. *Annals of nuclear energy*, 139:107250.
- [73] Lee, K., Jessee, M., Collins, B., Taylor, Z., and Graham, A. (2022). Transient convective delayed neutron precursors of ²³⁵u for the molten salt reactor experiment. Technical report, Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States).
- [74] Leppänen, J. (2007). Development of a New Monte Carlo Reactor Physics Code. *VTT Publications*, 640:241.
- [75] Leppänen, J., Pusa, M., Viitanen, T., Valtavirta, V., and Kaltiaisenaho, T. (2015). The Serpent Monte Carlo code: Status, development and applications in 2013. *Annals of Nuclear Energy*, 82:142–150.
- [76] Li, R., Wang, S., Rineiski, A., Zhang, D., and Merle-Lucotte, E. (2015). Transient analyses for a molten salt fast reactor with optimized core geometry. *Nuclear Engineering and Design*, 292:164–176.
- [77] Lindsay, A., Ridley, G., Rykhlevskii, A., and Huff, K. (2018). Introduction to Moltres: An application for simulation of Molten Salt Reactors. *Annals of Nuclear Energy*, 114:530–540.
- [78] Loaiza, D. J., Brunson, G., Sanchez, R., and Butterfield, K. (1998). Measurements of absolute delayed neutron yield and group constants in the fast fission of ²³⁵u and ²³⁷np. *Nuclear science and engineering*, 128(3):270–277.
- [79] Loaiza, D. J. and Haskin, F. E. (2000). Dominant delayed neutron precursors to model reactivity predictions for multiple fissioning nuclides. *Nuclear science and engineering*, 134(1):22–36.
- [80] Manero, F. and Konshin, V. (1972). Status of the energy-dependent average ν -values for the heavy isotopes ($Z > 90$) from thermal to 15 mev and of ν -values for spontaneous fission. *Atomic Energy Review*, 10(4):637–756.
- [81] Merle-Lucotte, E., Heuer, D., Allibert, M., Ghetta, V., Brun, C. L., Brissot, R., Liatard, E., and Mathieu, L. (2007). The thorium molten salt reactor: Launching the thorium cycle while closing the current fuel cycle. In *European Nuclear Conference*, page 7, Brussels, Belgium.
- [82] Merle-Lucotte, E., Heuer, D., Allibert, M., Ghetta, V., and Le Brun, C. (2008). Introduction to the physics of molten salt reactors. In *Materials Issues for Generation IV Systems: Status, Open Questions and Challenges*, pages 501–521. Springer.
- [83] Merle-Lucotte, E., Heuer, D., Laureau, A., Brovchenko, M., Allibert, M., Aufiero, M., and Rubiolo, P. (2015). Physical assessment of the load following and starting procedures for the molten salt fast reactor. In *International Congress on Advances in Nuclear Power Plants (ICAPP2015)*, number 15450, pages 901–909.
- [84] Meulekamp, R. K. and Van Der Marck, S. C. (2006). Calculating the effective delayed neutron fraction with monte carlo. *Nuclear science and engineering*, 152(2):142–148.
- [85] Mills, R., Dillmann, I., Dimitriou, P., Singh, B., Abriola, D., Garrett, P., Xiaolong, H., Fallot, M., Porta, A., Mukherjee, G., Chiba, S., Minato, F., Pksaykin, V., Ott, D., Tain, J., Rykaczewski, K., Sonzogni, A., Johnson, T., Semkova, V., and VerPELLI, M. (2014). Calculation of delayed neutron emission from fission and fitting results to few group approximations.
- [86] Nagy, K., Kloosterman, J. L., and Lathouwers, D. (2008). Parametric studies on the fuel salt composition in thermal molten salt breeder reactors. *International Conference on the Physics of Reactors*, page 8.

- [87] Nicolino, C., Lapenta, G., Dulla, S., and Ravetto, P. (2008). Coupled dynamics in the physics of molten salt reactors. *Annals of Nuclear Energy*, 35(2):314–322.
- [88] Nuttin, A., Heuer, D., Billebaud, A., Brissot, R., Brun, C. L., Liatard, E., Meplan, O., Merle-Lucotte, E., Nifenecker, H., and Perdu, F. (2005a). POTENTIAL OF THORIUM MOLTEN SALT REACTORS : DETAILED CALCULATIONS AND CONCEPT EVOLUTION WITH A VIEW TO LARGE SCALE ENERGY PRODUCTION. *Progress in Nuclear Energy*, 46(1):23.
- [89] Nuttin, A., Heuer, D., Billebaud, A., Brissot, R., Le Brun, C., Liatard, E., Loiseaux, J.-M., Mathieu, L., Meplan, O., Merle-Lucotte, E., et al. (2005b). Potential of thorium molten salt reactors: detailed calculations and concept evolution with a view to large scale energy production. *Progress in nuclear energy*, 46(1):77–99.
- [90] Ott, K. and Neuhold, R. (1985). *Introductory Nuclear Reactor Dynamics*. American Nuclear Society.
- [91] Parish, T., Charlton, W., Shinohara, N., Andoh, M., Brady, M., and Raman, S. (1999). Status of six-group delayed neutron data and relationships between delayed neutron parameters from the macroscopic and microscopic approaches. *Nuclear science and engineering*, 131(2):208–221.
- [92] Park, J., Jeong, Y., Lee, H. C., and Lee, D. (2015). Whole core analysis of molten salt breeder reactor with online fuel reprocessing: Whole core analysis of MSBR with online fuel reprocessing. *International Journal of Energy Research*, pages n/a–n/a.
- [93] Perry, R. T., Wilson, W. B., England, T. R., and Brady, M. C. (1986). Application of evaluated fission-product delayed-neutron precursor data in reactor kinetics calculations. *Radiation Effects*, 94(1-4):43–47.
- [94] Pfeiffer, B., Kratz, K.-L., and Möller, P. (2002). Status of delayed-neutron precursor data: half-lives and neutron emission probabilities. *Progress in nuclear energy*, 41(1-4):39–69.
- [95] Protection, P. (2011). Evaluation methodology for proliferation resistance and physical protection of generation iv nuclear energy systems.
- [96] Ridley, G. and Chvala, O. (2017). A method for predicting fuel maintenance in once-through MSRs. *Annals of Nuclear Energy*, 110:265–281.
- [97] Rineiski, A., Sinitsa, V., Maschek, W., and Wang, S. (2005). Kinetics and cross-section developments for analyses of reactor transmutation concepts with simmer. *Proc. Mathematics and Computation, Supercomputing, Reactor Physics and Nuclear and Biological Applications, Avignon, France*.
- [98] Roberts, R., Meyer, R., and Wang, P. (1939). Further observations on the splitting of uranium and thorium. *Physical Review*, 55(5):510.
- [99] Robertson, R. (1971). CONCEPTUAL DESIGN STUDY OF A SINGLE-FLUID MOLTEN-SALT BREEDER REACTOR. Technical Report ORNL-4541, 4030941, ORNL.
- [100] Robertson, R. C. (1965). Msre design & operations report part 1 description of reactor design. Technical report, Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States).
- [101] Rodriguez-Vieitez, E., Lowenthal, M., Greenspan, E., and Ahn, J. (2002). TRANSMUTATION CAPABILITY OF ONCE-THROUGH CRITICAL OR SUB-CRITICAL MOLTEN-SALT REACTORS. In *National Research Council*.
- [102] Romano, P. K., Horelik, N. E., Herman, B. R., Nelson, A. G., Forget, B., and Smith, K. (2015). OpenMC: A state-of-the-art Monte Carlo code for research and development. *Annals of Nuclear Energy*, 82:90–97.
- [103] Romero-Barrientos, J., Damián, J. M., Molina, E., Zambra, M., Aguilera, P., López-Usquiano, E., Parra, B., and Ruiz, A. (2022). Calculation of kinetic parameters β_{eff} and λ with modified open source monte carlo code openmc (td). *Nuclear Engineering and Technology*, 54(3):811–816.
- [104] Rudstam, G. (1982). Six-group representation of the energy spectra of delayed neutrons from fission. *Nuclear Science and Engineering*, 80(2):238–255.

- [105] Rudstam, G., Finck, P., Filip, A., D'angelo, A., and McKnight, R. (2002). Delayed neutron data for the major actinides. *A report by the Working Party on International Evaluation Co-operation of the OECD NEA Nuclear Science Committee*, 6.
- [106] Rykhlevskii, A., Bae, J. W., and Huff, K. D. (2019). Modeling and simulation of online reprocessing in the thorium-fueled molten salt breeder reactor. *Annals of Nuclear Energy*, 128:366–379.
- [107] Seifert, L. (2023). *Analysis of and comparison between reprocessing methods in the molten salt breeder reactor*. PhD thesis, University of Illinois at Urbana-Champaign.
- [108] Shahbazi, S., Romano, P., Fei, T., and Grabaskas, D. (2022). Steady-state radiochemical transport model of the molten salt reactor experiment. *Journal of Radioanalytical and Nuclear Chemistry*, 331(12):5247–5257. 4.
- [109] Shalev, S. and Rudstam, G. (1972). Delayed neutron emission from ^{137}I . *Physical Review Letters*, 28(11):687.
- [110] Sheu, R., Chang, C., Chao, C., and Liu, Y.-W. (2013). Depletion analysis on long-term operation of the conceptual Molten Salt Actinide Recycler & Transmuter (MOSART) by using a special sequence based on SCALE6/TRITON. *Annals of Nuclear Energy*, 53:1–8.
- [111] Shi, C., Cheng, M., and Liu, G. (2016). Development and application of a system analysis code for liquid fueled molten salt reactors based on relap5 code. *Nuclear Engineering and Design*, 305:378–388.
- [112] Shi, J. and Fratoni, M. (2021). Gen-foam model and benchmark of delayed neutron precursor drift in the molten salt reactor experiment. In *EPJ Web of Conferences*, volume 247, page 06040. EDP Sciences.
- [113] Suzuki, N. and Shimazu, Y. (2008). Reactivity-initiated-accident analysis without scram of a molten salt reactor. *Journal of nuclear science and technology*, 45(6):575–581.
- [114] Tano, M., Walker, S. A., and Abou-Jaoude, A. (2023a). A Coupled Griffin and Pronghorn Method for Spatially Resolved Depletion Analysis of Molten Salt Reactors. In *American Nuclear Society Mathematics & Computation*.
- [115] Tano, M., Walker, S. A., and Abou-Jaoude, A. (2023b). A Coupled Griffin and Pronghorn Method for Spatially Resolved Depletion Analysis of Molten Salt Reactors. In *American Nuclear Society Mathematics & Computation*.
- [116] The pandas development team (2024). pandas-dev/pandas: Pandas.
- [117] Tuttle, R. (1975). Delayed-neutron data for reactor-physics analysis. *Nuclear Science and Engineering*, 56(1):37–71.
- [118] Van Rossum, G. and Drake, F. L. (2009). *Python 3 Reference Manual*. CreateSpace, Scotts Valley, CA.
- [119] Verboomen, B., Haeck, W., and Baeten, P. (2006). Monte carlo calculation of the effective neutron generation time. *Annals of nuclear energy*, 33(10):911–916.
- [120] Vergnes, J. and Lecarpentier, D. (2002). The amster concept (actinides molten salt transmuter). *Nuclear engineering and design*, 216(1-3):43–67.
- [121] Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., et al. (2020a). Scipy 1.0: fundamental algorithms for scientific computing in python. *Nature methods*, 17(3):261–272.
- [122] Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., Carey, C. J., Polat, İ., Feng, Y., Moore, E. W., VanderPlas, J., Laxalde, D., Perktold, J., Cimrman, R., Henriksen, I., Quintero, E. A., Harris, C. R., Archibald, A. M., Ribeiro, A. H., Pedregosa, F., van Mulbregt, P., and SciPy 1.0 Contributors (2020b). SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, 17:261–272.

- [123] Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., Carey, C. J., Polat, İ., Feng, Y., Moore, E. W., VanderPlas, J., Laxalde, D., Perktold, J., Cimrman, R., Henriksen, I., Quintero, E. A., Harris, C. R., Archibald, A. M., Ribeiro, A. H., Pedregosa, F., van Mulbregt, P., and SciPy 1.0 Contributors (2020c). SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, 17:261–272.
- [124] Waldo, R., Karam, R., and Meyer, R. (1981). Delayed neutron yields: Time dependent measurements and a predictive model. *Physical Review C*, 23(3):1113.
- [125] Wilson, W. and England, T. (2002). Delayed neutron study using endf/b-vi basic nuclear data. *Progress in Nuclear Energy*, 41(1-4):71–107.
- [126] Wooten, D. and Powers, J. J. (2018). A review of molten salt reactor kinetics models. *Nuclear Science and Engineering*, 191(3):203–230.
- [127] Xia, S., Chen, J., Guo, W., Cui, D., Han, J., Wu, J., and Cai, X. (2019). Development of a Molten Salt Reactor specific depletion code MODEC. *Annals of Nuclear Energy*, 124:88–97.
- [128] Yamamoto, T., Mitachi, K., Ikeuchi, K., and Suzuki, T. (2006). Transient characteristics of small molten salt reactor during blockage accident. *Heat Transfer—Asian Research: Co-sponsored by the Society of Chemical Engineers of Japan and the Heat Transfer Division of ASME*, 35(6):434–450.
- [129] Zanetti, M., Cammi, A., Luzzi, L., Krepel, J., Mikityuk, K., et al. (2015). Extension of the fast code system for the modelling and simulation of msr dynamics. In *Proceedings of ICAPP*, pages 03–06.
- [130] Zhang, D., Qiu, S., Su, G., Liu, C., and Qian, L. (2009). Analysis on the neutron kinetics for a molten salt reactor. *Progress in nuclear energy*, 51(4-5):624–636.
- [131] Zhou, S., Yang, W. S., Park, T., and Wu, H. (2018). Fuel cycle analysis of molten salt reactors based on coupled neutronics and thermal-hydraulics calculations. *Annals of Nuclear Energy*, 114:369–383.
- [132] Zhuang, K., Li, T., Zhang, Q., He, Q., and Zhang, T. (2020). Extended development of a Monte Carlo code OpenMC for fuel cycle simulation of molten salt reactor. *Progress in Nuclear Energy*, 118:103115.
- [133] Zuo, X.-D., Cheng, M.-S., Dai, Y.-Q., Yu, K.-C., and Dai, Z.-M. (2022). Flow field effect of delayed neutron precursors in liquid-fueled molten salt reactors. *Nuclear Science and Techniques*, 33(8):96.